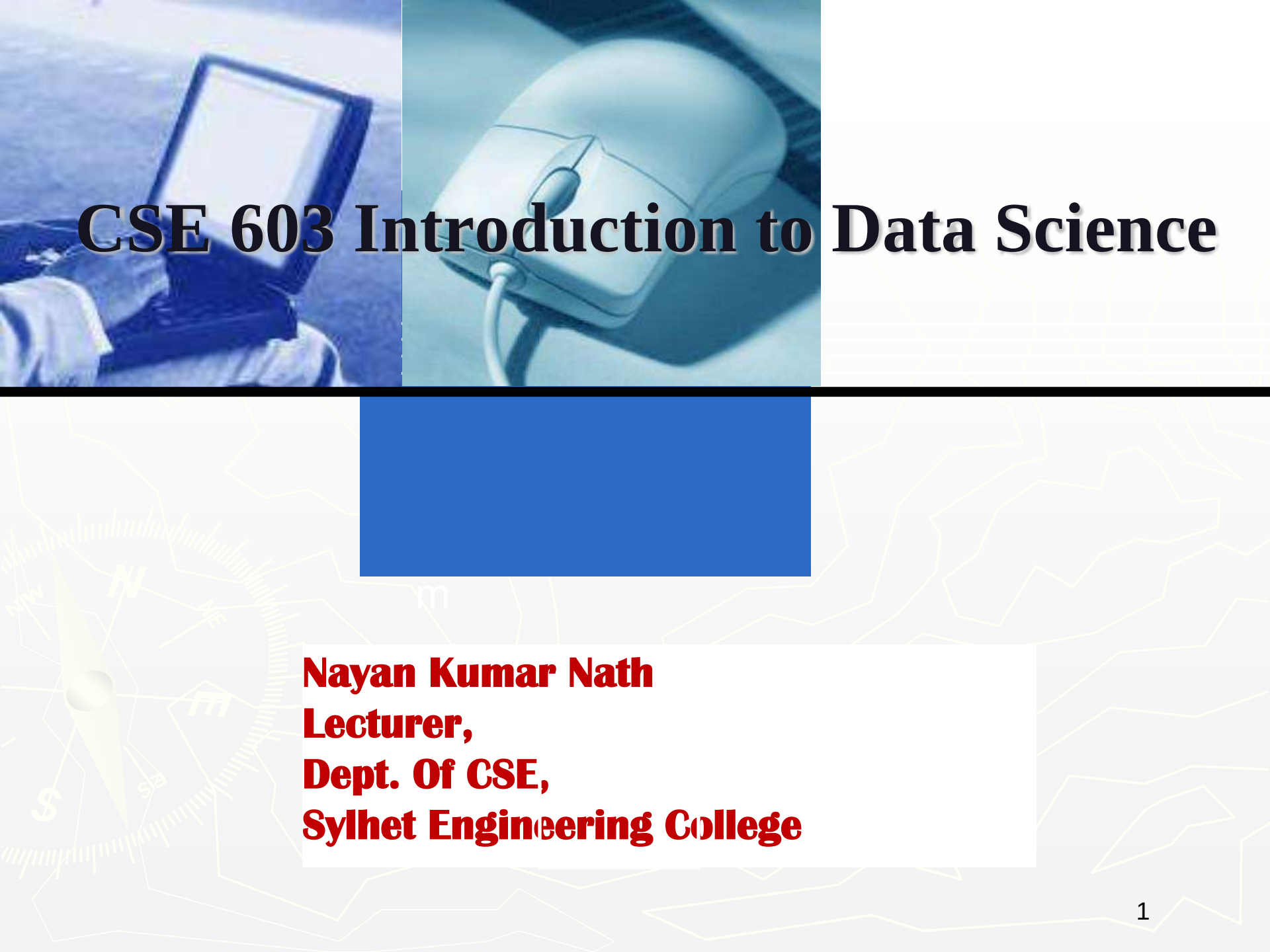
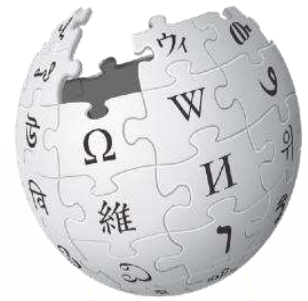




CSE 603 Introduction to Data Science

Nayan Kumar Nath
Lecturer,
Dept. Of CSE,
Sylhet Engineering College

WHAT IS DATA SCIENCE?

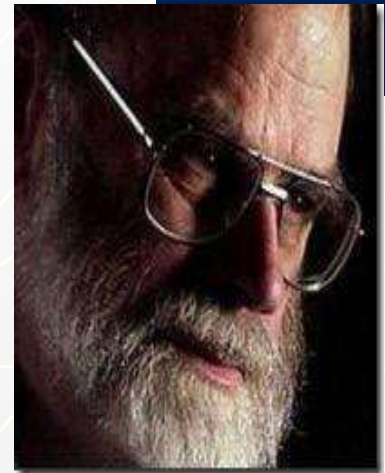
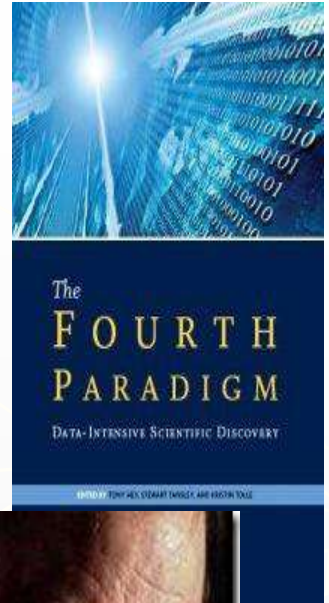


WIKIPEDIA
The Free Encyclopedia

- “**Data science**, also known as **data-driven science**, is an interdisciplinary field of scientific methods, processes, algorithms and systems to extract knowledge or insights from data in various forms, either structured or unstructured, similar to data mining.”
- Data science is the field of study that combines domain expertise, programming skills, and knowledge of math and statistics to extract meaningful insights from data. Data science practitioners apply machine learning algorithms to numbers, text, images, video, audio, and more to produce **artificial intelligence (AI)** systems that perform tasks which ordinarily require human intelligence. In turn, these systems generate insights that analysts and business users translate into tangible business value. (**Data Robot**)

WHAT IS DATA SCIENCE?

- Fourth paradigm
 - “... change of all sciences moving from observational, to theoretical, to computational and now to the 4th Paradigm – Data-Intensive Scientific Discovery”



What is Data Science?

Despite the hype, useful definitions exist¹

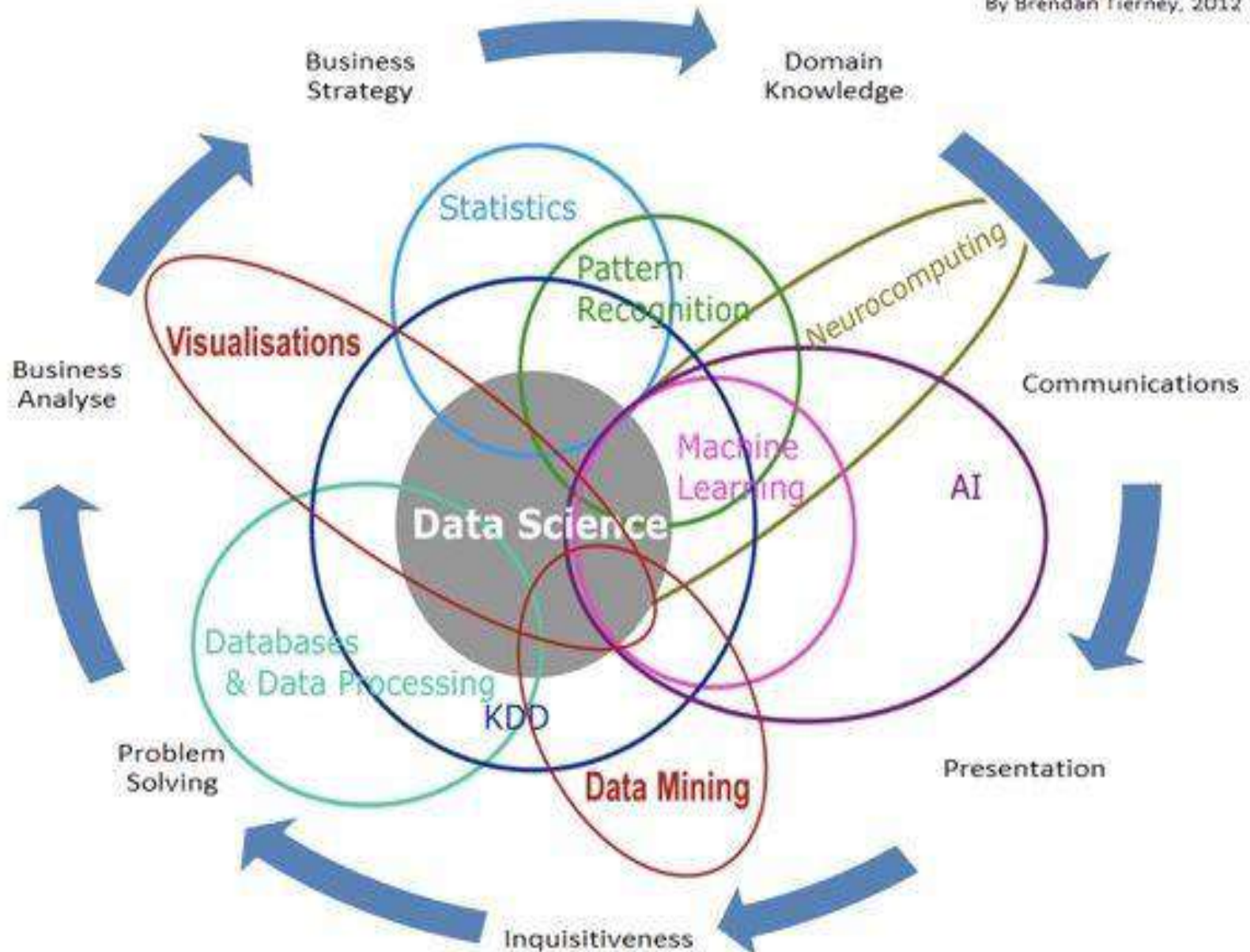
*Data science is about the extraction of **useful** information and knowledge from **large** volumes of data, in order to **improve** business **decision-making***

Is an **interdisciplinary** subject with 3 key areas:

- Statistics
- Computer science
- Domain expertise

Data Science Is Multidisciplinary

By Brendan Tierney, 2012



So Data Science is?

Data Science is apply Data with Science

-Bern Jonathan



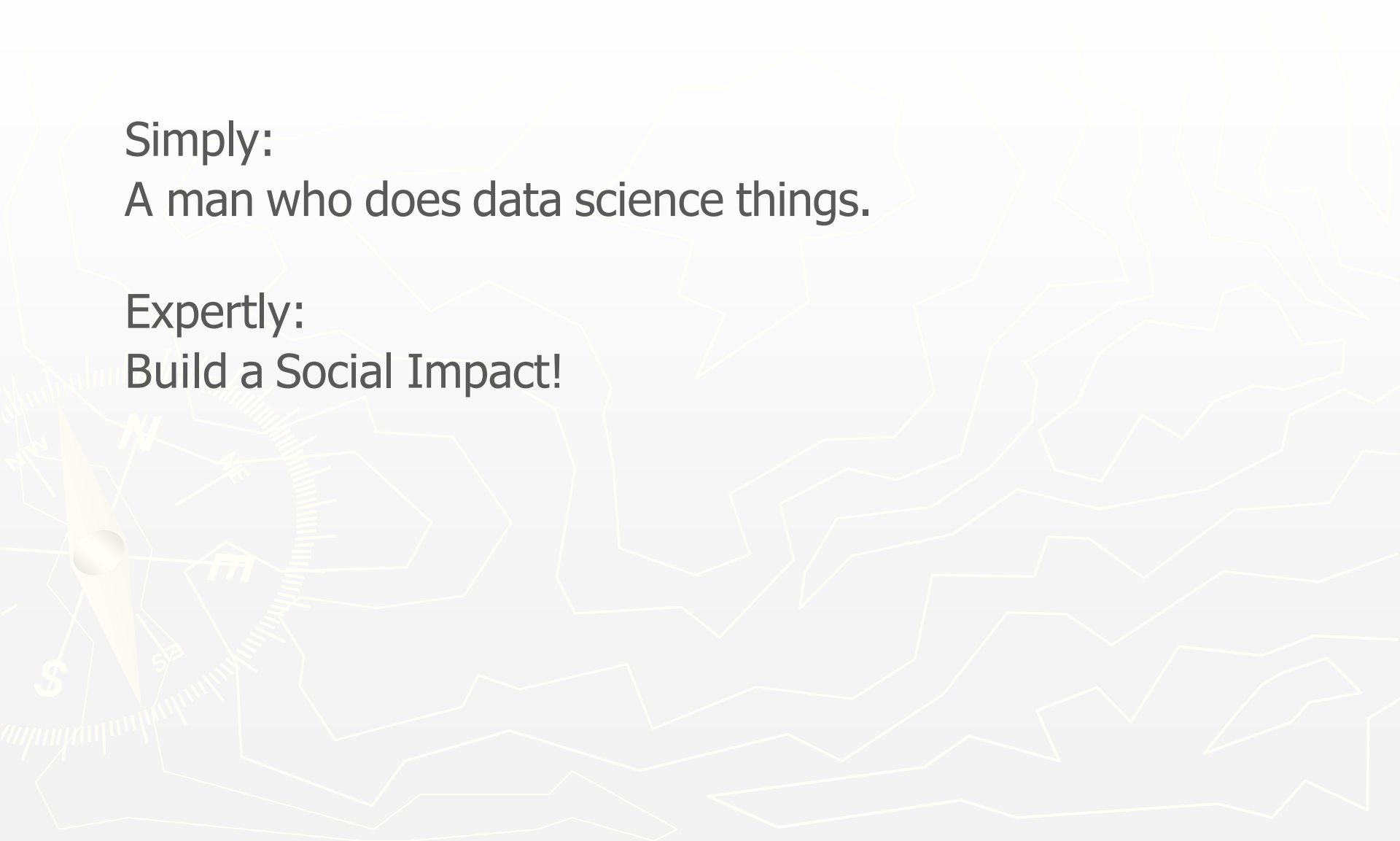
Who is Data Scientist?

Simply:

A man who does data science things.

Expertly:

Build a Social Impact!



Data Science?

DATA SCIENTIST

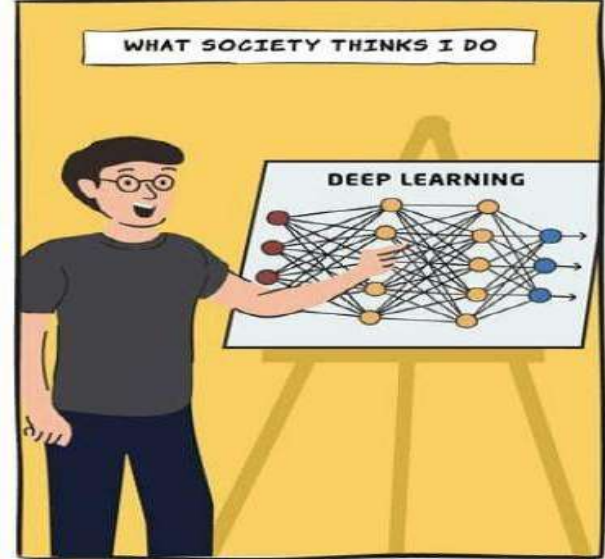
WHAT MY FRIENDS THINK I DO



WHAT MY MOM THINKS I DO



WHAT SOCIETY THINKS I DO



WHAT MY BOSS THINKS I DO



WHAT I THINK I DO



WHAT I ACTUALLY DO





(((Josh Wills)))

@josh_wills



Follow

Data Scientist (n.): Person who is better at statistics than any software engineer and better at software engineering than any statistician.

RETWEETS

1,506

LIKES

1,087



12:55 AM - 4 May 2012



48



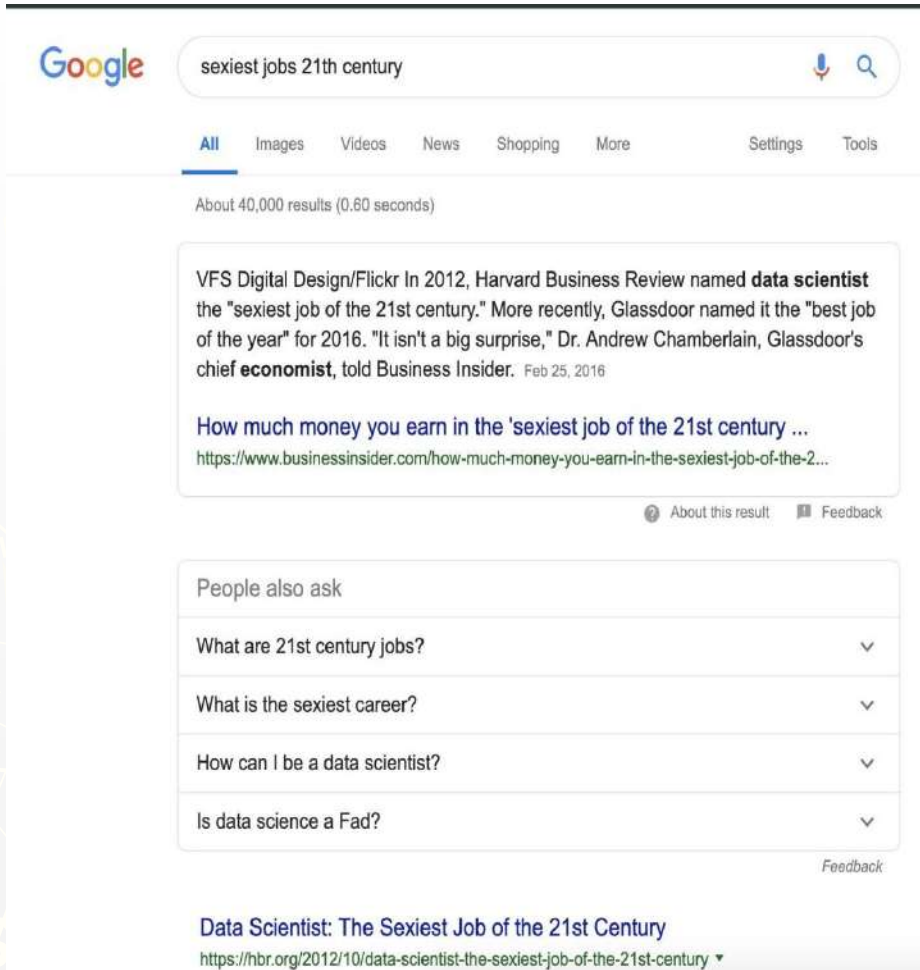
1.5K



1.1K



Data Scientist



Google

sexiest jobs 21th century

All Images Videos News Shopping More Settings Tools

About 40,000 results (0.60 seconds)

VFS Digital Design/Flickr In 2012, Harvard Business Review named **data scientist** the "sexiest job of the 21st century." More recently, Glassdoor named it the "best job of the year" for 2016. "It isn't a big surprise," Dr. Andrew Chamberlain, Glassdoor's chief **economist**, told Business Insider. Feb 25, 2016

[How much money you earn in the 'sexiest job of the 21st century ...](#)
<https://www.businessinsider.com/how-much-money-you-earn-in-the-sexiest-job-of-the-2...>

About this result Feedback

People also ask

- What are 21st century jobs? ▾
- What is the sexiest career? ▾
- How can I be a data scientist? ▾
- Is data science a Fad? ▾

Feedback

[Data Scientist: The Sexiest Job of the 21st Century](#)
<https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century> ▾

Data Team

Data Scientist

also known as Data Managers, statisticians.



A data scientist will be able to take data science projects from end to end. They can help store large amounts of data, create predictive modelling processes and present the findings.

Skills: Mathematics, Programming, Communication



Will use programmes such as:
SQL, Python, R

Data Engineers

also known as database administrators and data architects.



They are versatile generalists who use computer science to help process large datasets. They typically focus on coding, cleaning up data sets, and implementing requests that come from data scientists.

Skills: Programming, Mathematics, Big data



Will use programmes such as:
Hadoop, NoSQL, and Python

Data Analysts

also known as business Analysts.



They typically help people from across the company understand specific queries with charts.

Skills: Statistics, Communication, Business knowledge



Will use programmes such as:
Excel, Tableau, SQL

What Data Science do?

A typical data science process looks like this, which can be modified for specific use case:

- Understand the business
- Collect & explore the data
- Prepare & process the data
- Build & validate the models
- Deploy & monitor the performance

Industry roles vs research roles of Data science

Industry Roles in Data Science:

These roles are typically focused on applying data science techniques to solve real-world business problems and improve operational efficiency.

Data Scientist:

Focus: Solve business problems using statistical models, machine learning, and deep learning techniques.

Tools: Python, R, SQL, TensorFlow, Scikit-learn, Hadoop, Spark.

Responsibilities: Data collection, cleaning, analysis, and predictive modeling. Deliver insights to inform business strategies.

Data Analyst:

Focus: Analyze data to generate actionable insights for business decisions.

Tools: Excel, SQL, Tableau, Power BI, Python (for more advanced analysis).

Responsibilities: Data visualization, reporting, identifying trends and patterns in data, supporting decision-making processes.

Machine Learning Engineer:

Focus: Implement machine learning models and algorithms into production.

Tools: Python, TensorFlow, Keras, PyTorch, cloud platforms like AWS or Azure.

Responsibilities: Writing scalable code, optimizing algorithms, deploying machine learning models, and maintaining production systems.

Data Engineer:

Focus: Build and manage the infrastructure and architecture for collecting, storing, and analyzing large datasets.

Tools: SQL, Hadoop, Spark, Python, data warehousing tools.

Responsibilities: Designing and maintaining data pipelines, ensuring data quality and accessibility for analysis, managing cloud databases.

Business Intelligence Analyst/Developer:

Focus: Extract business insights through data integration and visualization.

Tools: Power BI, Tableau, SQL, Excel, and ETL tools.

Responsibilities: Creating dashboards and reports, defining metrics, analyzing data trends to guide business decisions.

AI Specialist:

Focus: Build and deploy AI systems, often using machine learning techniques.

Tools: Python, TensorFlow, Keras, natural language processing (NLP), deep learning libraries.

Responsibilities: Developing AI applications, enhancing user experience, improving automation processes.

Industry roles vs research roles of Data science

Research Roles in Data Science:

Research roles in Data Science are typically focused on developing new techniques, models, and methodologies for data analysis that could be applied across various industries.

Research Scientist (Data Science):

Focus: Conduct fundamental research to create new algorithms or methods in areas like machine learning, AI, or statistics.

Tools: Python, R, Jupyter notebooks, advanced statistical tools, cloud computing.

Responsibilities: Publishing research papers, proposing new methodologies, experimenting with novel techniques, collaborating with academic institutions.

Quantitative Researcher:

Focus: Work with complex mathematical models to identify patterns in financial markets or other areas.

Tools: Python, R, MATLAB, specialized quantitative research software.

Responsibilities: Developing quantitative models, optimizing algorithms, using statistical methods to derive insights, testing hypotheses.

Machine Learning Researcher:

Focus: Develop new machine learning models or improve existing algorithms in terms of accuracy, efficiency, or applicability.

Tools: Python, TensorFlow, PyTorch, Keras, Jupyter notebooks, academic libraries.

Responsibilities: Exploring new learning techniques, publishing research papers, collaborating with academic institutions.

AI Research Scientist:

Focus: Pioneering research in the field of Artificial Intelligence, including deep learning, reinforcement learning, and neural networks.

Tools: Deep learning libraries (TensorFlow, PyTorch), NLP libraries (spaCy, NLTK), mathematical models.

Responsibilities: Conducting experiments, publishing new findings, advancing AI models, and testing theories.

Data Science Researcher in Healthcare:

Focus: Apply data science techniques to medical and healthcare data to discover new insights, improve patient care, and develop predictive models.

Tools: Python, R, SQL, bioinformatics tools, machine learning models.

Responsibilities: Conducting research studies in healthcare, publishing papers, improving healthcare practices, developing new diagnostic tools.

Academia/Teaching:

Focus: Teaching and mentoring students, guiding research, and contributing to the academic community by publishing research papers.

Tools: Research software, teaching tools, academic libraries.

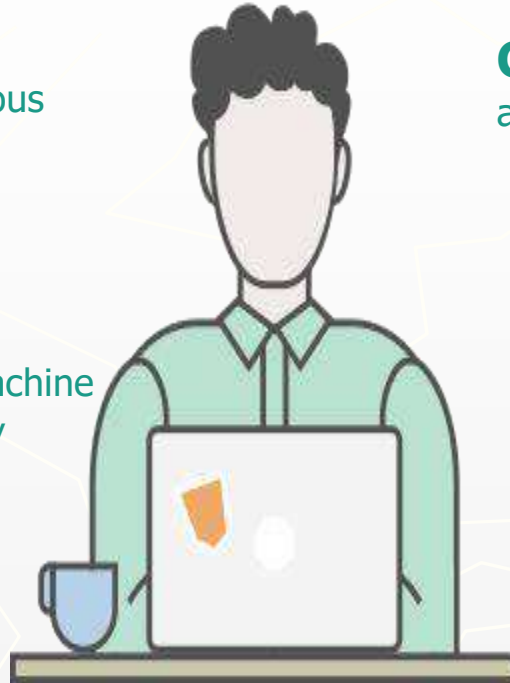
Responsibilities: Designing curriculums, mentoring students, conducting studies, and publishing academic papers on various data science topics.

WHAT DOES A GOOD DATA SCIENTIST LOOK LIKE?

Inquistive – skeptical and curious

Knowledgeable – knows machine learning, statistics, and probability

Scientific Method – Creates hypotheses, tests them, and updates understanding



Coding – is good at coding, hacking, and general programming

Product Oriented – knows how to build data products and visualizations to make data understandable to mere mortals

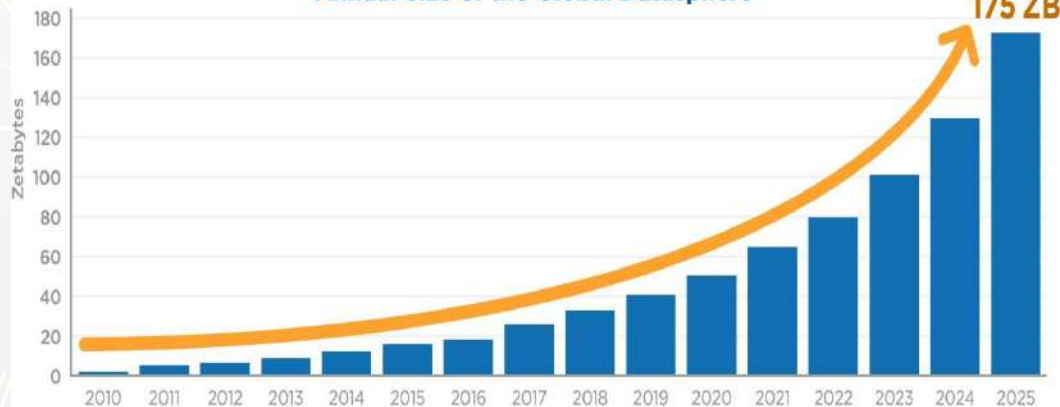
Domain Knowledge – understands the business and how to tell the relevant story from business data. Able to find answers to *known unknowns*.

Why is Data Science Important?

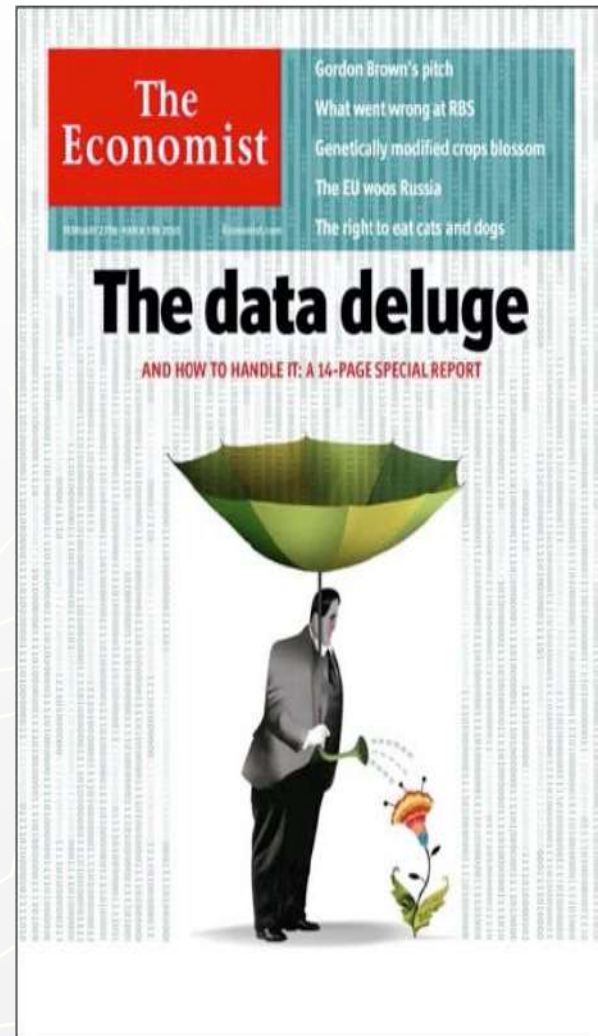
In the past, data analysis was typically **slow**:
needed teams of statisticians, analysts etc to
explore data **manually**

Today: **volume**, **velocity**, and **variety** make
manual analysis **impossible** ...

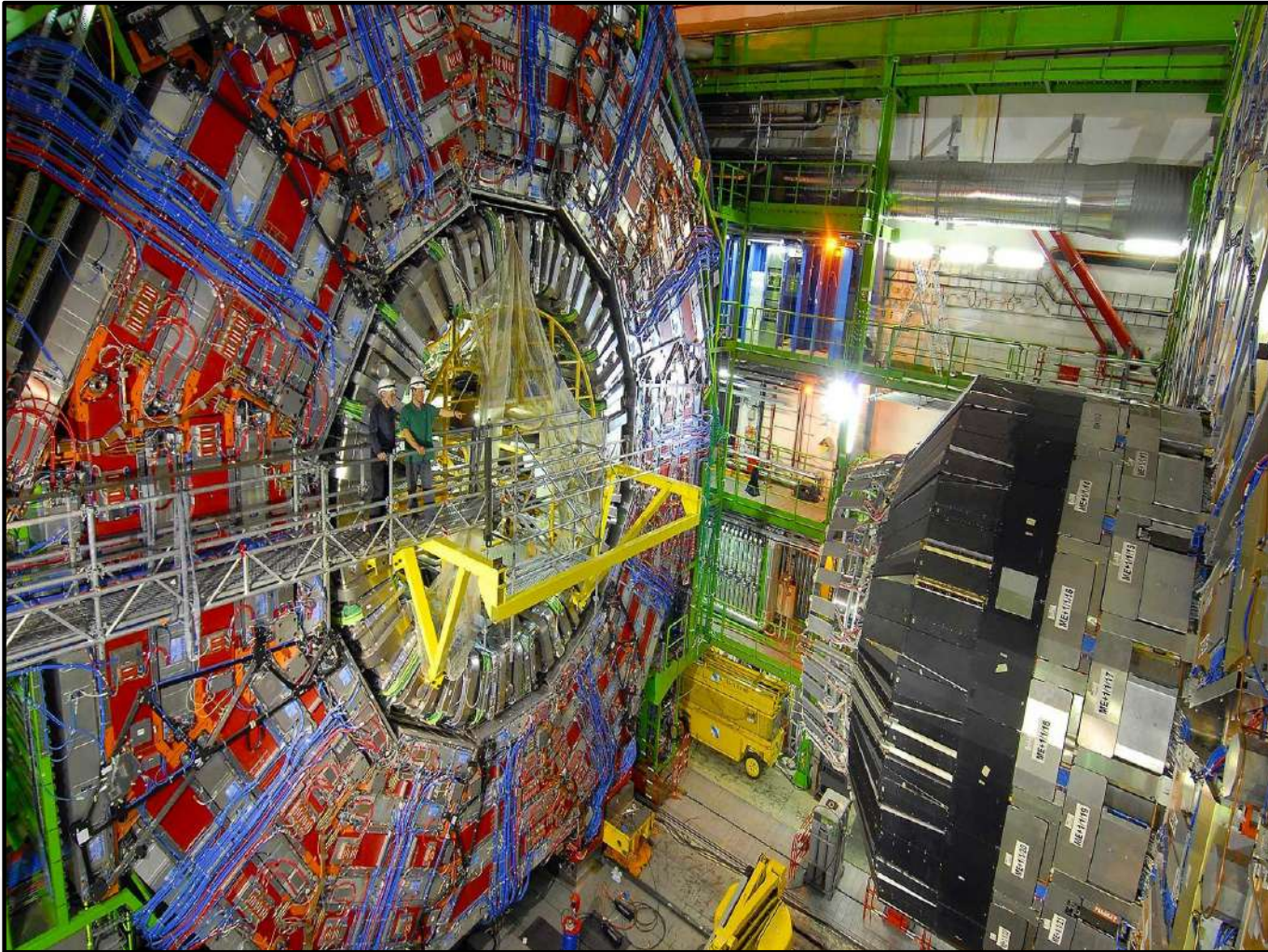
Annual Size of the Global Datasphere



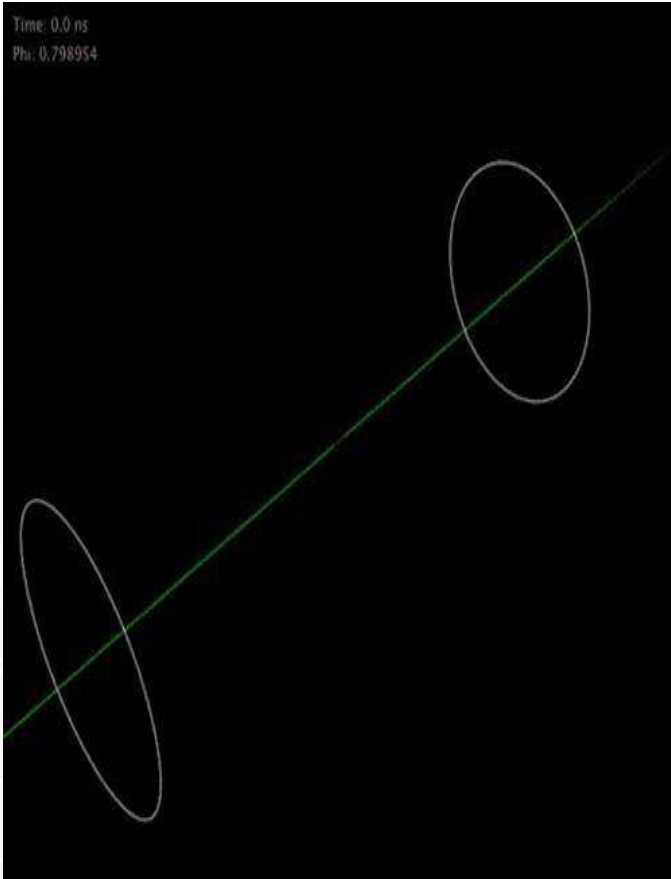
Source: Data Age 2025, sponsored by Seagate with data from IDC Global DataSphere, Nov 2018



Big Data: The Large Hadron Collider at CERN



Big Data: The Large Hadron Collider at CERN



- **150 million sensors** delivering data 40 million times per second.
- There are nearly **600 million collisions** per second.
- Only **100 collisions of interest** per second.
- Raw data production exceeds **500 exabytes** per day (1 EB = 1 million TB).
- Due to **filtering** only 200 petabyte are generated annually (1 PB = 1000 TB).

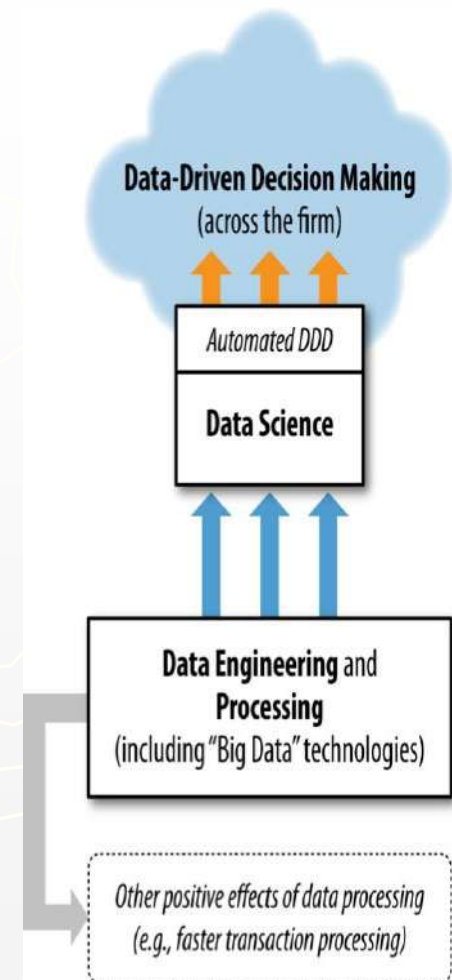
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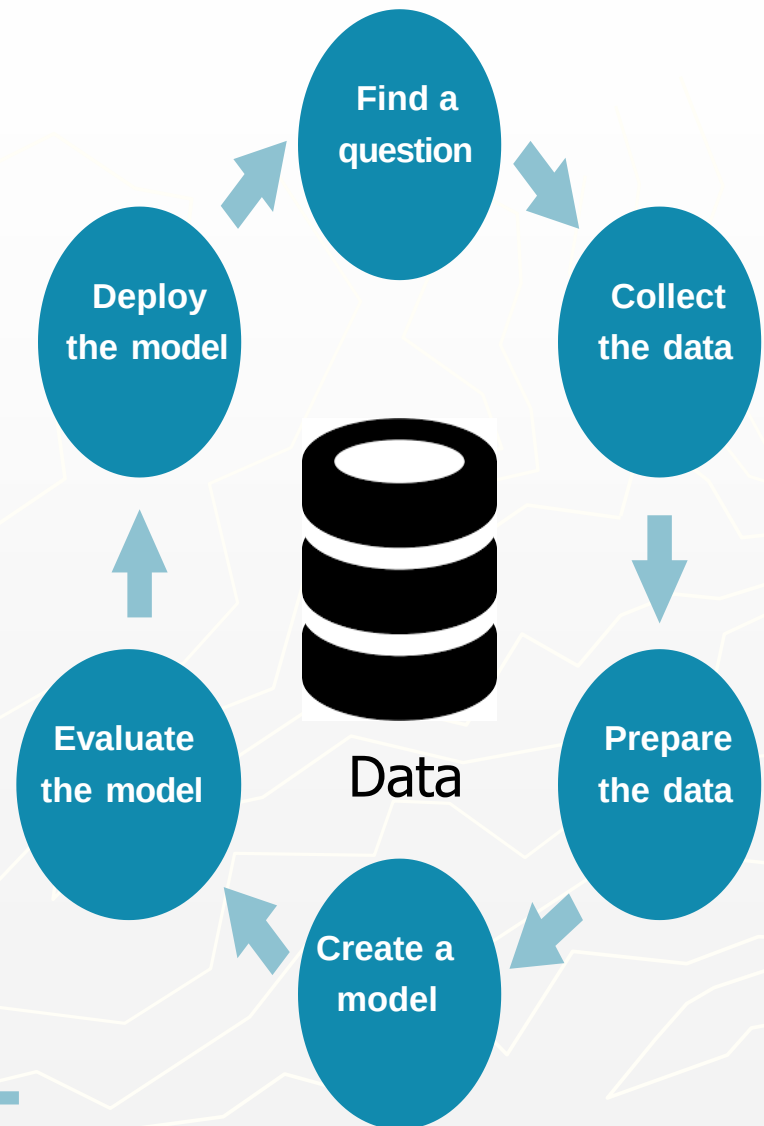
... but **fast computers** and **good algorithms**
allow much deeper analyses than before

-) **data-driven** decision making
-) base decisions on analysis of data,
not intuition



The Data Science Process

How is data science performed?



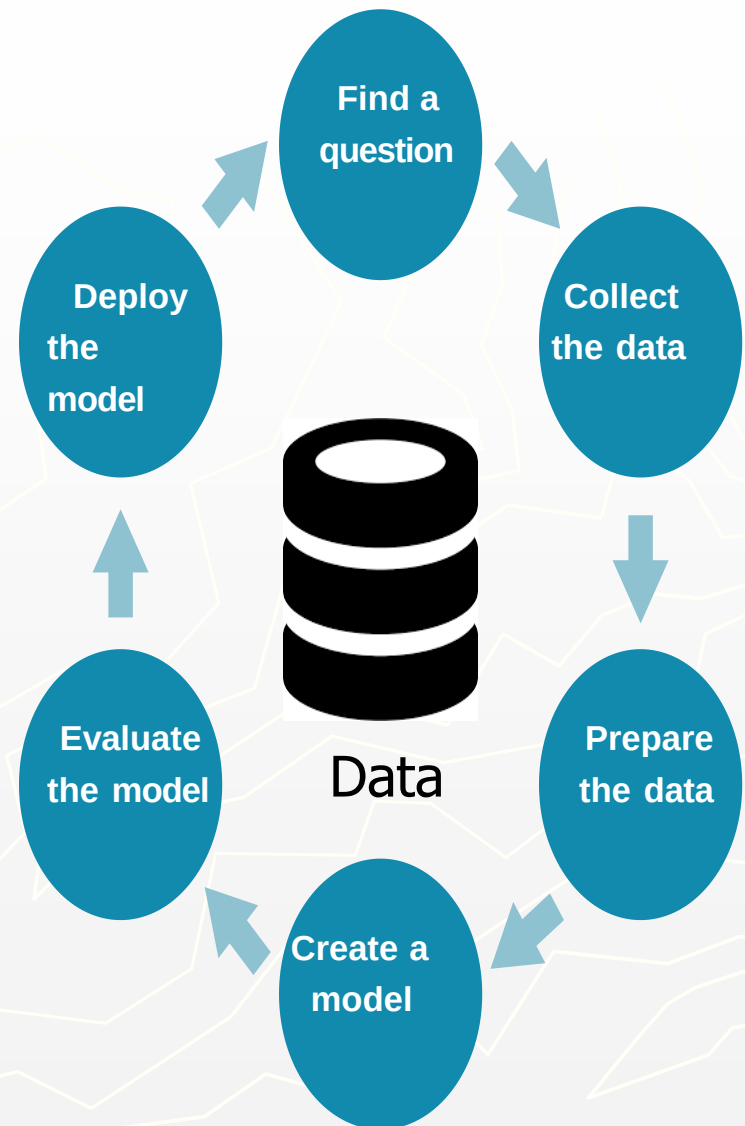
The Data Science Process

How is data science performed?

- Iterative process
- Nonsequential
- Early termination
- Established

processes, e.g. CRISP-DM

(<https://bit.ly/1tX658>)



Example # 1

Hurricane Frances was on its way, barreling across the Caribbean, threatening a direct hit on Florida's Atlantic coast ... A week ahead of the storm's landfall, Linda M. Dillman, Wal- Mart's chief information officer, pressed her staff to come up with forecasts based on what had happened when Hurricane Charley struck several weeks earlier.¹



Why might data-driven predictions be useful in this scenario?

¹What Wal-Mart Knows About Customer Habits, NYT (2004)

Example # 1

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7x increase in
sales before
hurricane



Top-selling
item was
beer!

Why might data-driven predictions be useful in this scenario?

¹What Wal-Mart knows about customer habits, NYT (2004)

Example # 2



"If we wanted to figure out if a customer is pregnant, even if she didn't want us to know, can you do that?"¹

Why might Target want to know when you're pregnant?

¹How Companies Learn Your Secrets, NYT (2012)

Example #2



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"If we wanted to figure out if a customer is pregnant, even if she didn't want us to know, can you do that?"¹



"My daughter got this in the mail! She's still in high school ...
Are you trying to encourage her to get pregnant?"

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Example #2



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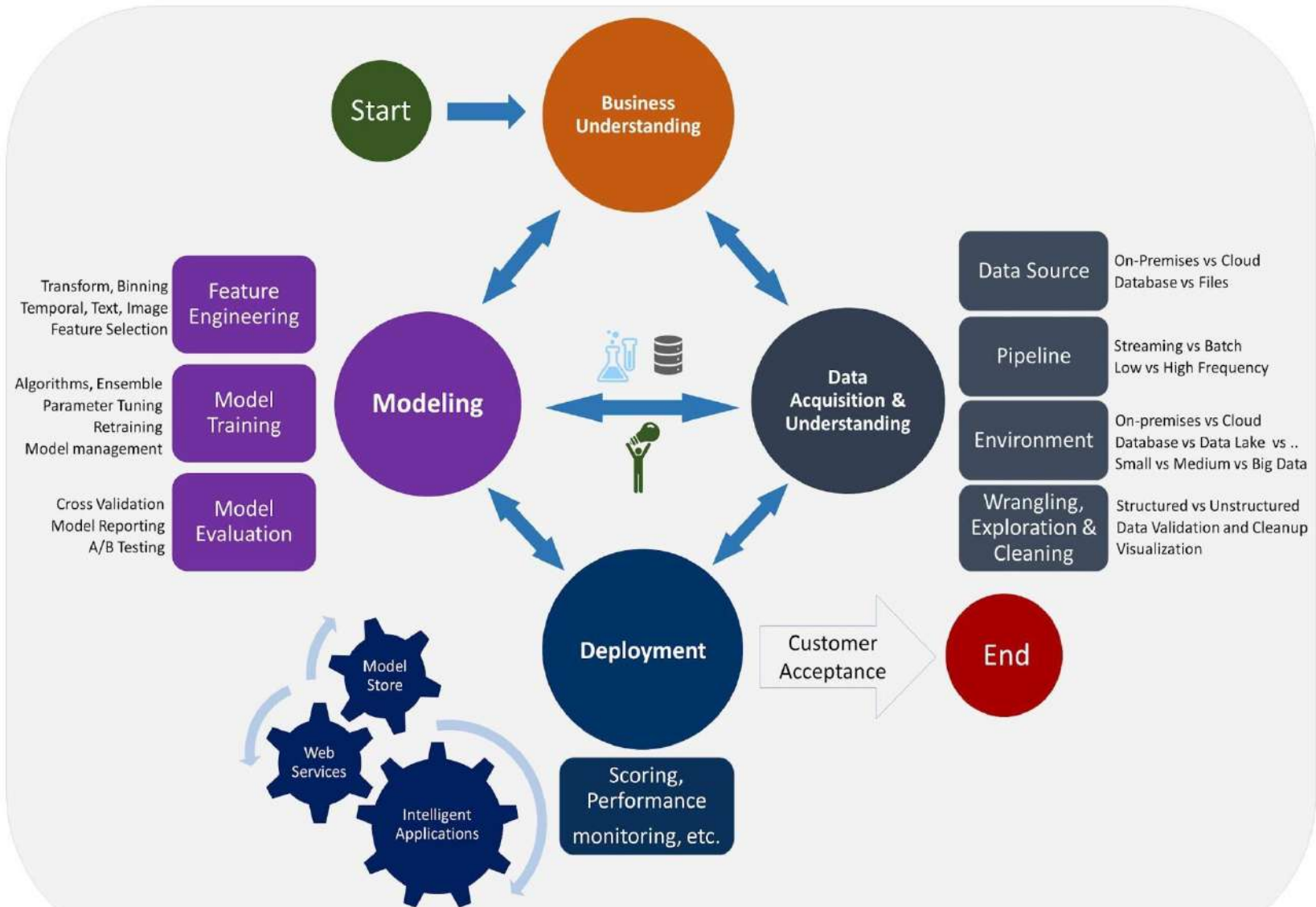
3 days later ...



"It turns out there's been some activities in my house I haven't been completely aware of. She's due in August. I owe you an apology."

¹How Companies Learn Your Secrets, NYT (2012)

Data Science Lifecycle



What's in the DataScience Methods

Statistical Methods

Sentiment analysis

Time series analysis

Data mining

Monte Carlo Optimization

Deep Learning

Multilevel modeling

Missing data imputations

Collaborative Filtering

Survival analysis

NLP

Pattern recognition

Classification and clustering

AB testing

Reinforcement Learning

Principal component and factor analysis

Logistic, multinomial and multiple linear regression techniques

Forecasting

Knowledge Graph

Propensity score matching

Network analysis



DATA SCIENCE APPLICATION EXAMPLES

- **Fraud detection**
 - Investigate fraud patterns in past data
 - Early detection is important
 - Before damage propagates
 - Harder than late detection
 - Precision is important
 - False positive and false negative are both bad
 - Real-time analytics



DATA SCIENCE APPLICATION EXAMPLES

- **Recommender systems**

- The ability to offer unique personalized service
- Increase sales, click-through rates, conversions, ...
 - Netflix recommender system valued at \$1B per year
 - Amazon recommender system drives a 20-35% lift in sales annually
- Collaborative filtering at scale



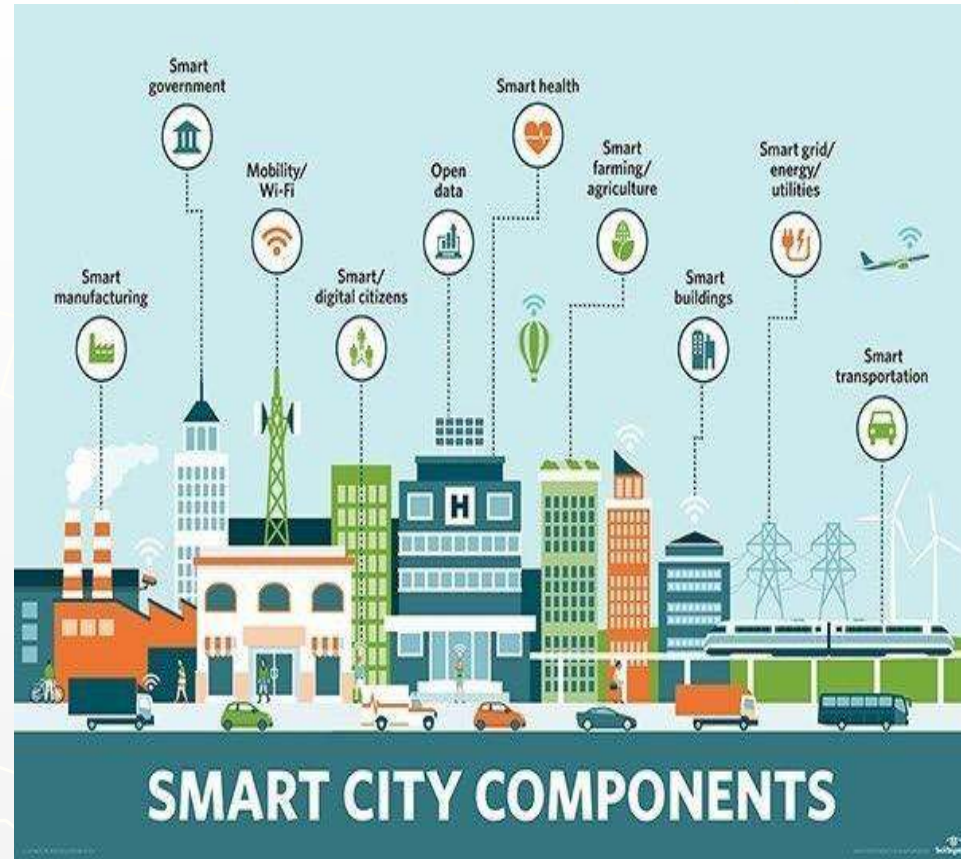
DATA SCIENCE APPLICATION EXAMPLES

- Predicting why patients are being readmitted
 - Reduce costs
 - Improve population health
 - Find the “why” behind specific populations being readmitted
 - Data lakes of multiple data sources
 - Investigate ties between readmission an socioeconomic data points, patient history, genetics, ...



DATA SCIENCE APPLICATION EXAMPLES

- “Smart cities”
 - Not well-defined



DATA SCIENCE APPLICATION EXAMPLES

- “Smart cities”
 - Not well-defined



lillispography / Getty / Emily Jan / The Atlantic

Stop Saying 'Smart Cities'

Digital stardust won't magically make future cities more affordable or resilient.

BRUCE STERLING

FEB 12, 2018

TECHNOLOGY

DATA SCIENCE APPLICATION EXAMPLES

- “Smart cities”
 - Not well-defined
 - Generally refers to using data and ICT to
 - Better plan communities
 - Better manage assets
 - Reduce costs
 - Deploy open data to better engage with community



illioisphotography / Getty / Emily Jan / The Atlantic

Stop Saying 'Smart Cities'

Digital stardust won't magically make future cities more affordable or resilient.

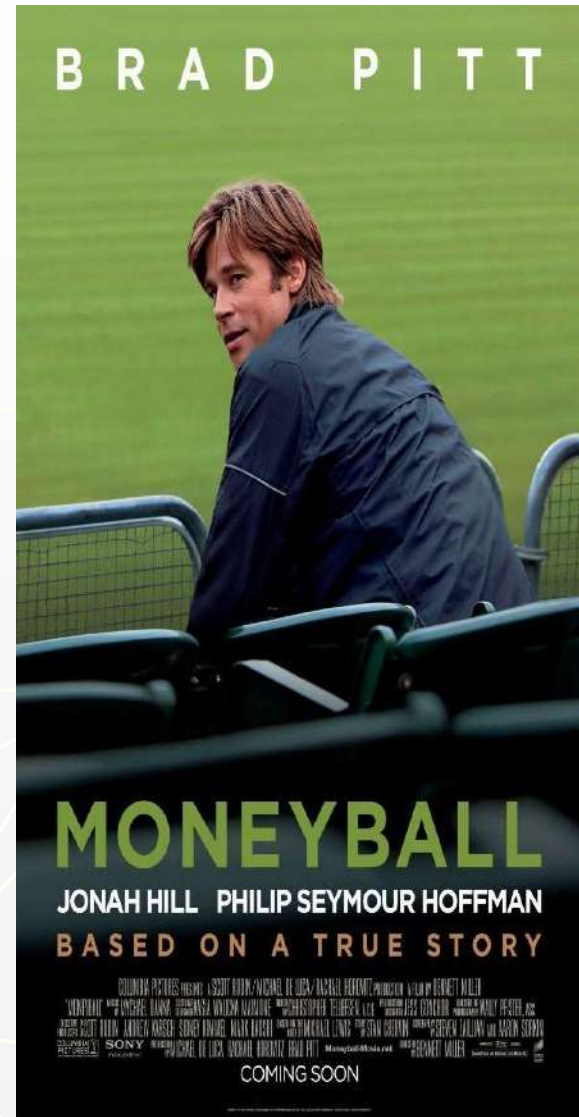
BRUCE STERLING

FEB 12, 2018

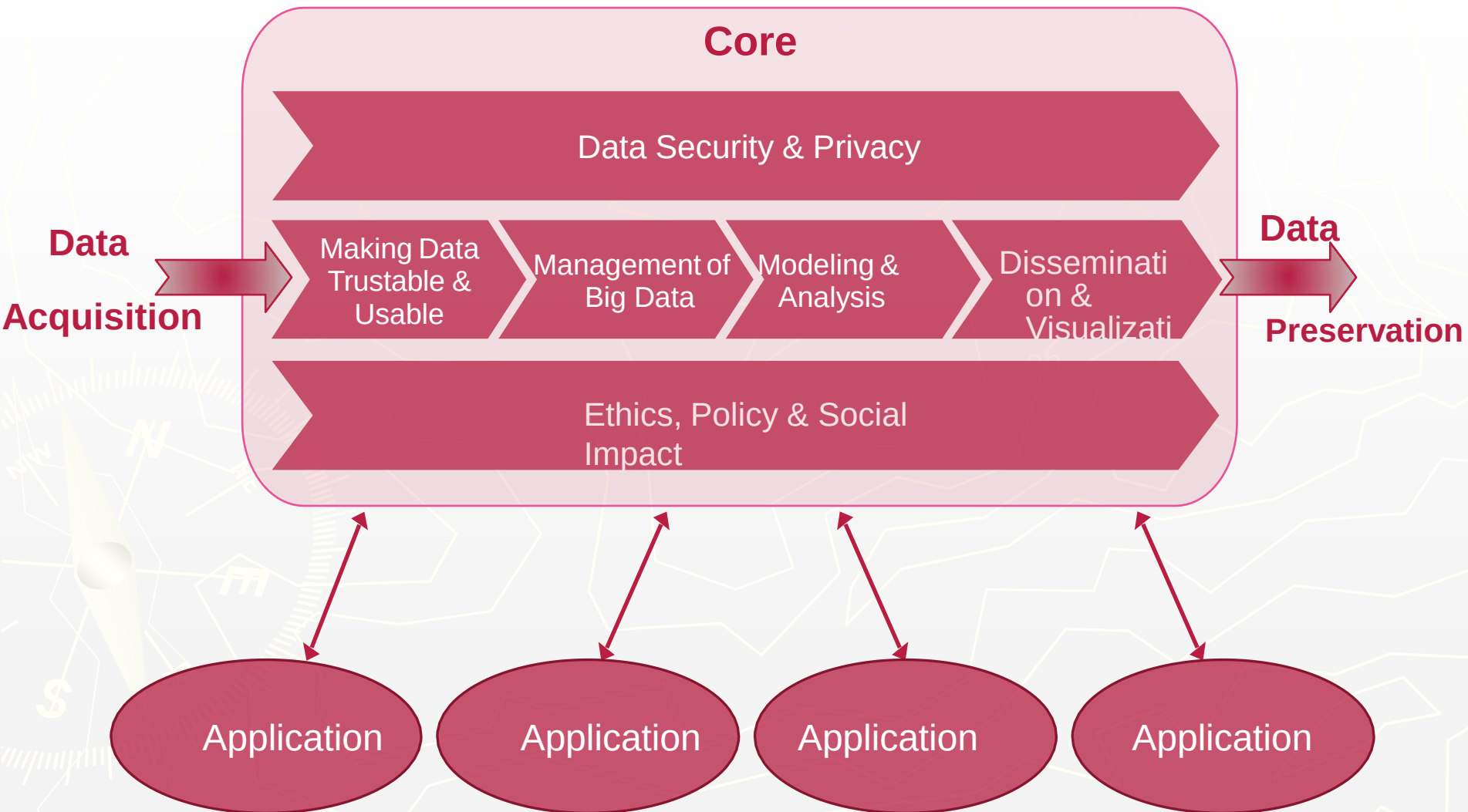
TECHNOLOGY

DATA SCIENCE APPLICATION EXAMPLES

- Moneyball
 - How to build a baseball team on a very low budget by relying on data
 - *Sabermetrics*: the statistical analysis of baseball data to objectively evaluate performance
 - 2002 record of 103-59 was **joint best** in MLB
 - Team salary budget: \$40 million
 - Other team: Yankees
 - Team salary budget: \$120 million



HOLISTIC APPROACH TO DATA SCIENCE



Commonly in Data Science

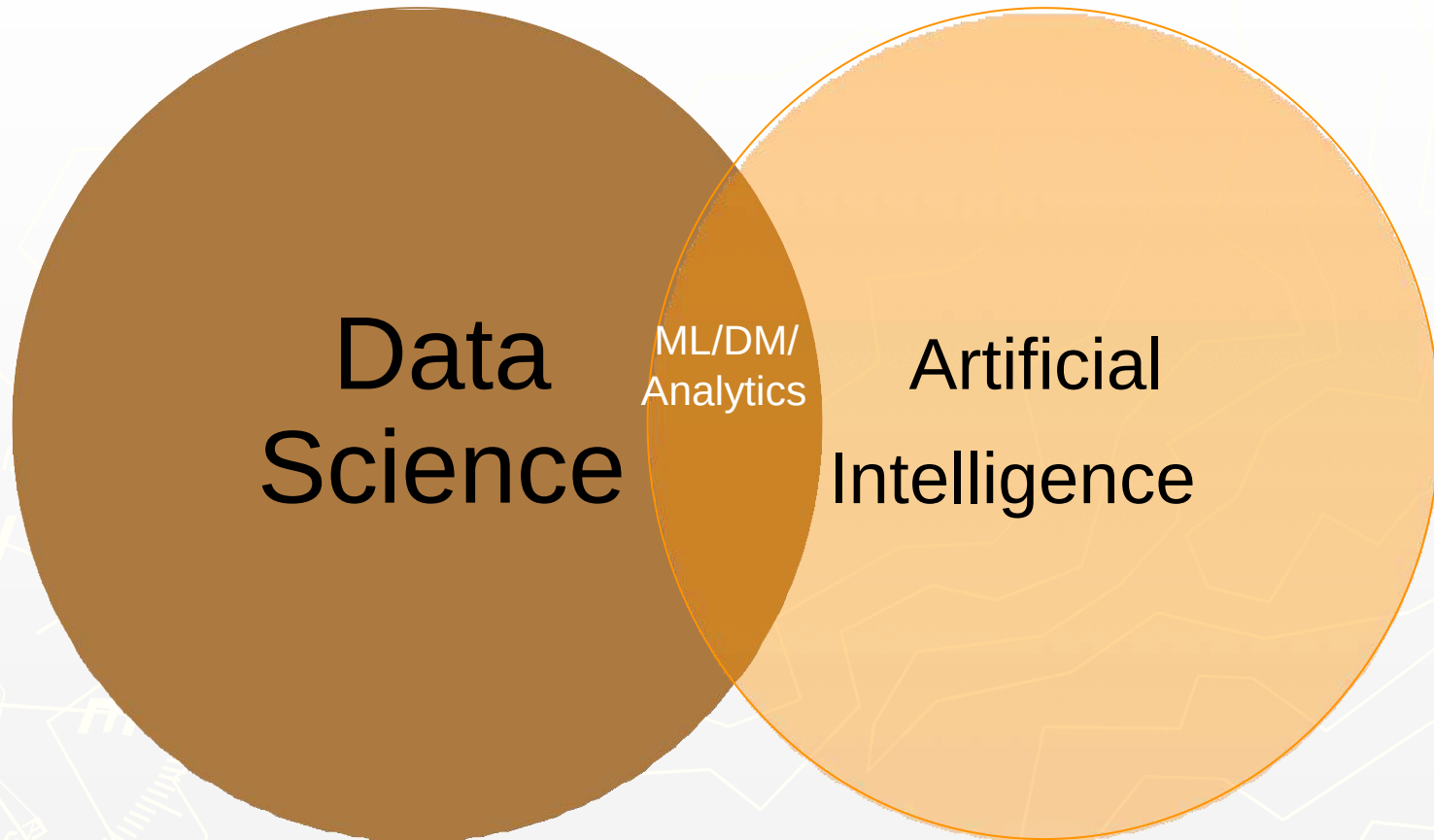
- **Supervised Learning (Classification & Regression)**
- **Unsupervised Learning (Clustering & Anomaly Detection)**
- **Reinforcement Learning**
- **Deep Learning**

DATA SCIENCE AND BIG DATA

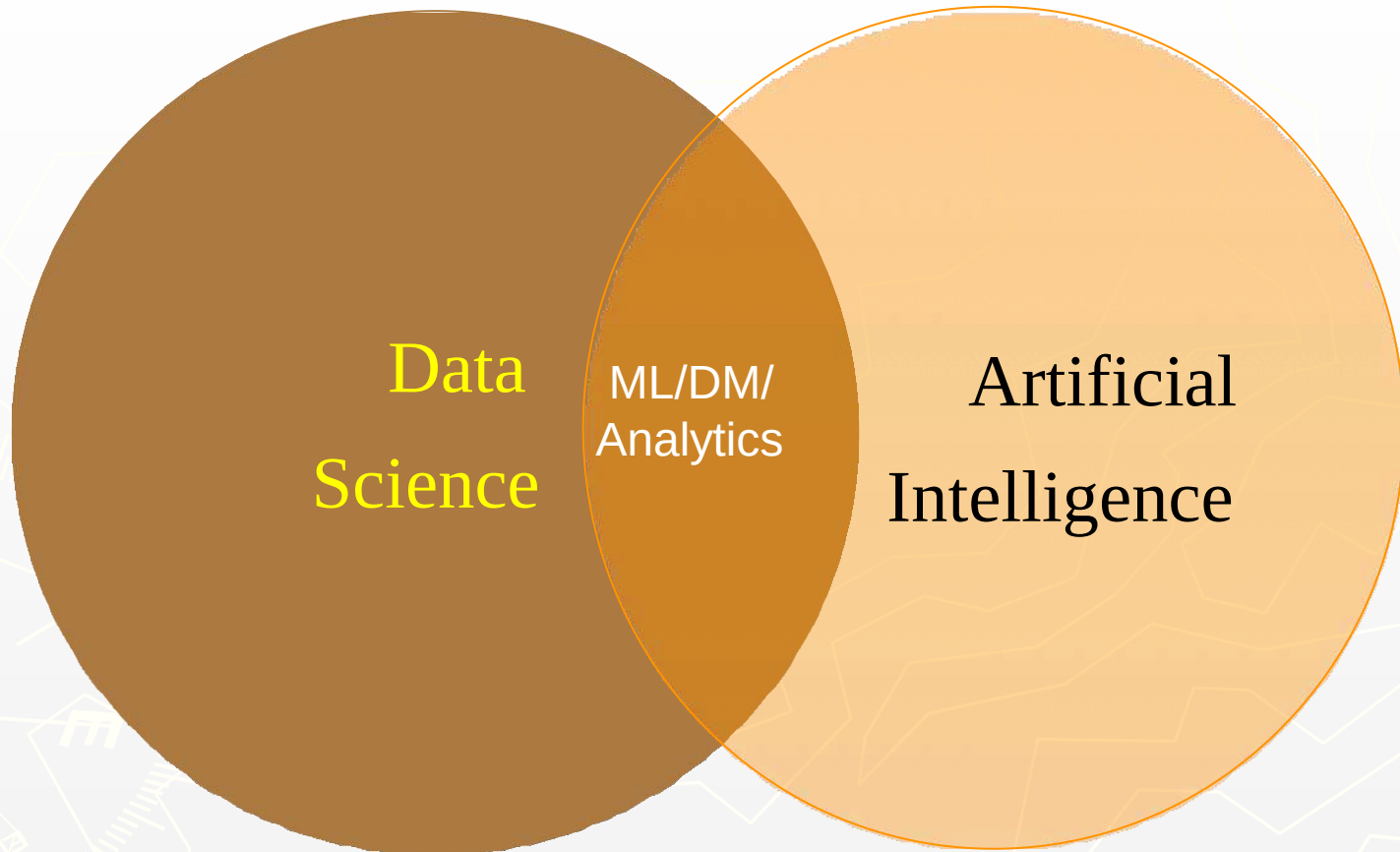
- They are not the “same thing”
- Big data = crude oil
 - Big data is about extracting “crude oil”, transporting it in “mega tankers”, siphoning it through “pipelines”, and storing it in “massive silos”
- Data science is about refining the “crude oil”

Carlos
Samohano Founder,
Data Science London

DATA SCIENCE AND ARTIFICIAL INTELLIGENCE



DATA SCIENCE AND ARTIFICIAL INTELLIGENCE



“**Data science** produces
insights.”

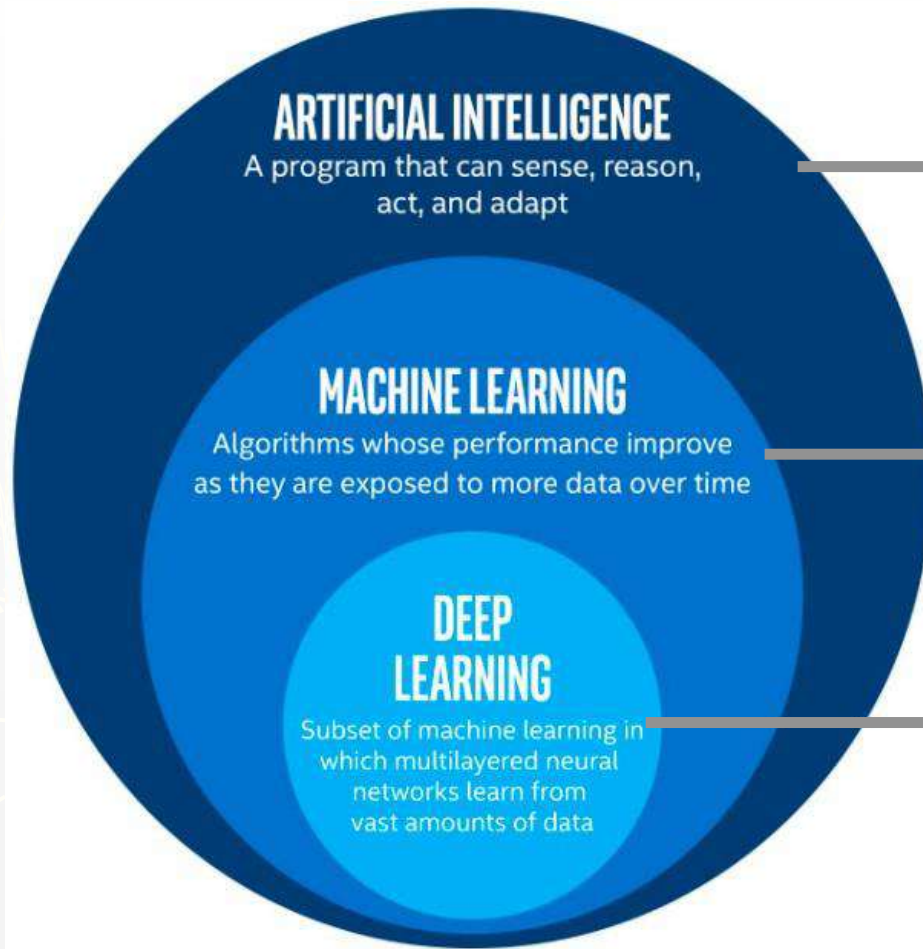
Machine learning produces
predictions”

What is machine learning?



Blue faces seem to be important...

What is machine learning really?

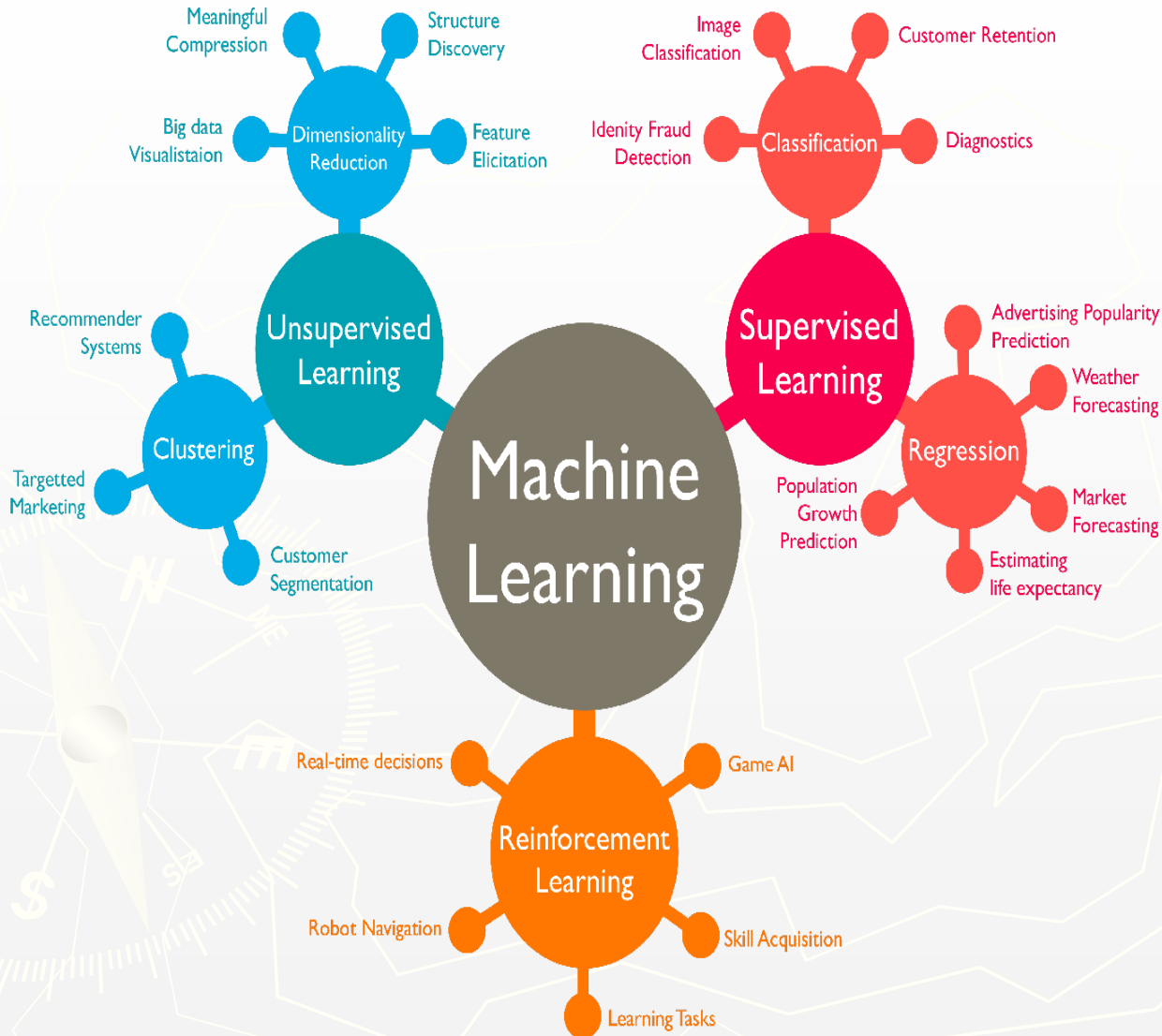


1950's: creation of first “intelligent” algorithms and programs. Alan Turing published “Computer Machinery and Intelligence” which proposed a test of machine intelligence called The Imitation Game

1980's: statistical models and algorithms that can learn from data

2010's: statistical models and algorithms inspired by neurones that can learn from data

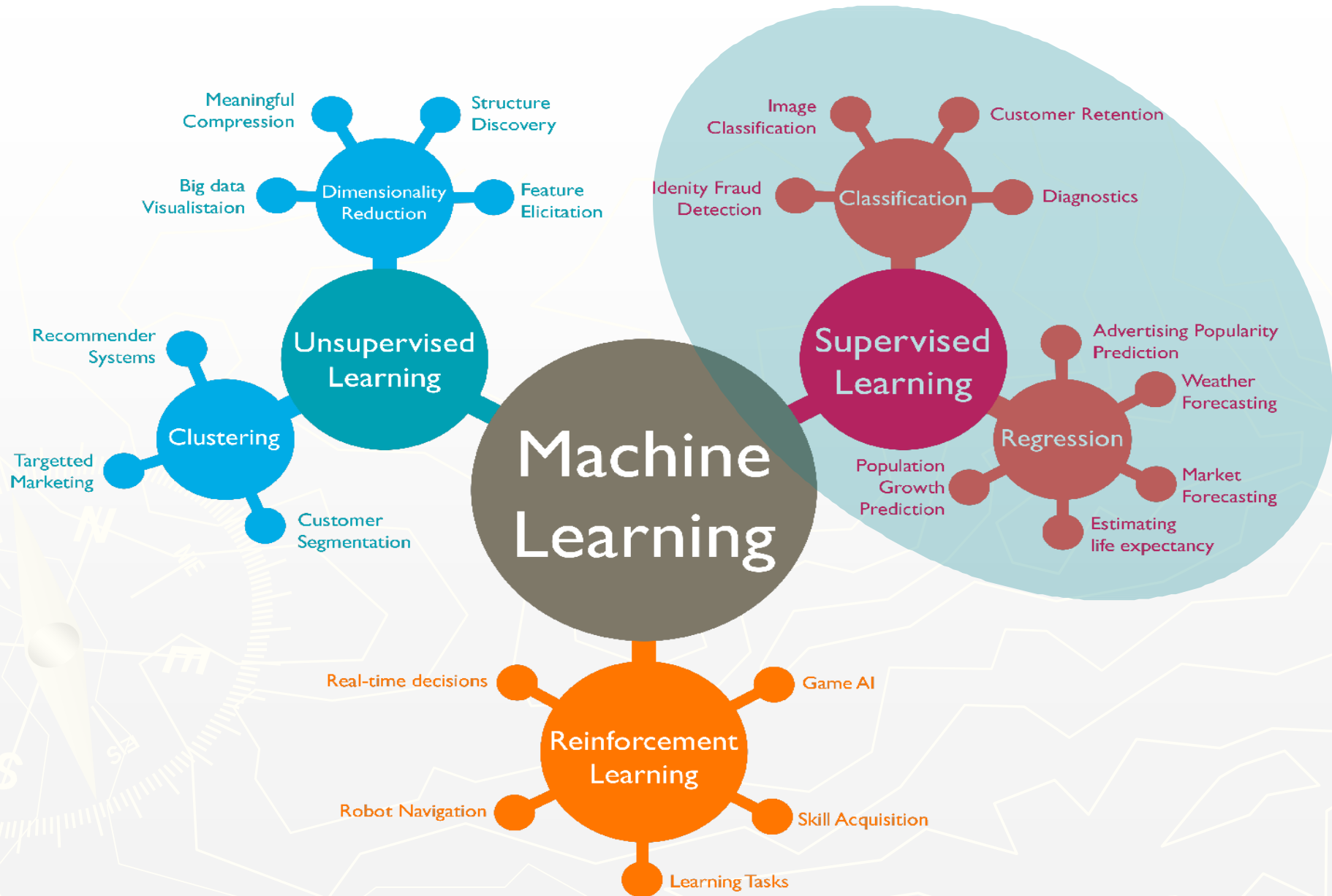
Machine Learning Branches



3 Main Branches:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Machine Learning Branches



Machine Learning Branches: Supervised Learning

In **supervised learning** the training data consists of **input/output pairs** and we train a function to map the inputs to the outputs.

Values, Vectors,
Words, Images etc.

Input

Classification

Regression

Categorical Variable
eg. Image Classification

A, B, C
Dogs/Cats

Continuous Variables
eg. Market Prediction

Prize/Cost,
Weight, Lifetime

Supervised Learning: Classification

Classification: Classification algorithms are used to solve the classification problems in which the output variable is categorical, such as **"Yes" or No, Male or Female, Red or Blue, etc.** The classification algorithms predict the categories present in the dataset. Assign categorical labels from a fixed set of labels to data samples. Some popular classification algorithms are given below:

Random Forest Algorithm, Decision Tree Algorithm, Logistic Regression Algorithm, Support Vector Machine Algorithm

Input
Data:

Output/Label:



**"Normal
Bike"**

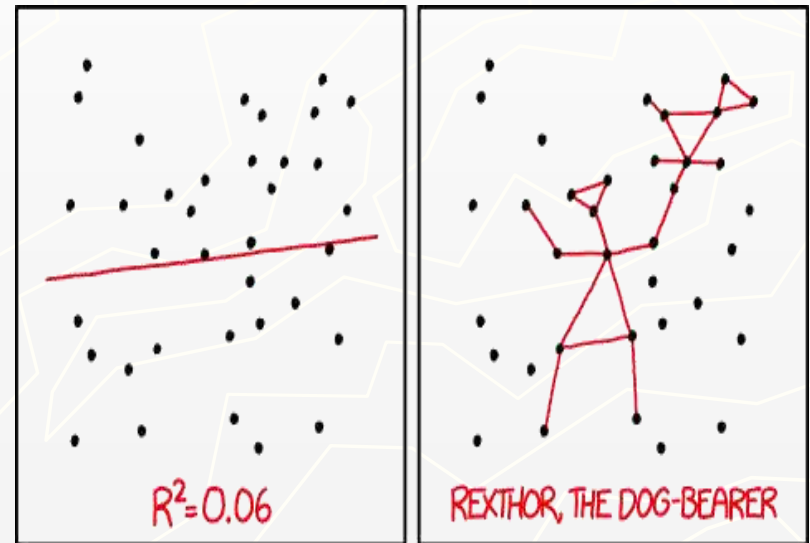
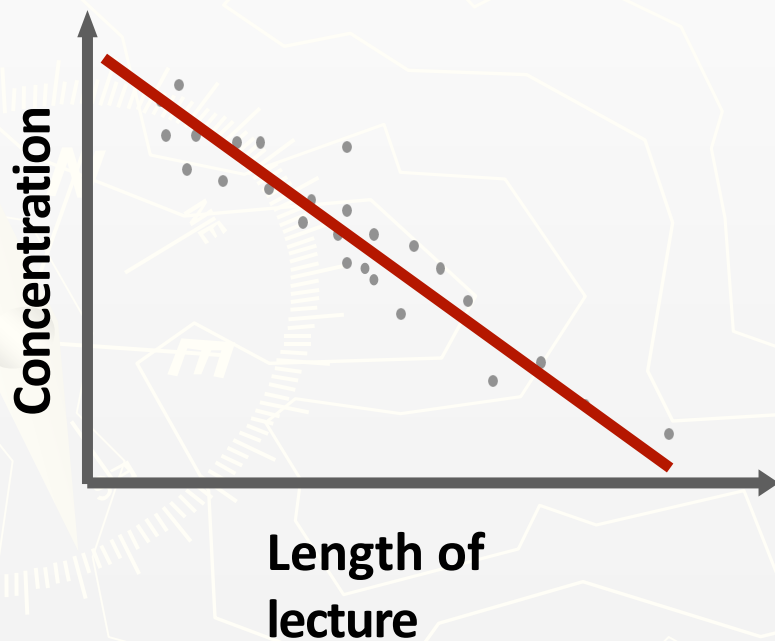


**"Broken
Bike"**

Supervised Learning: Regression

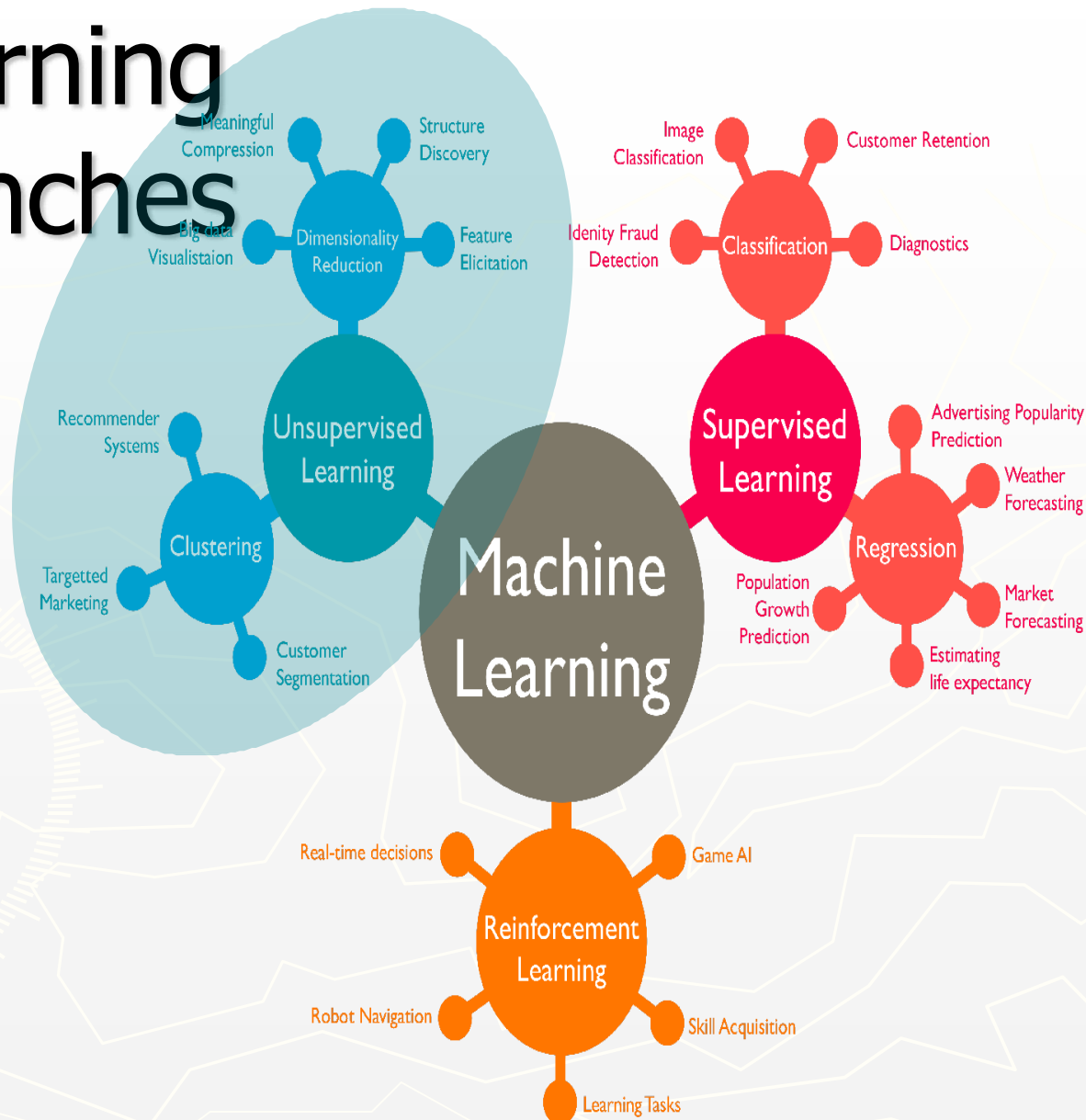
Regression: Regression algorithms are used to solve regression problems in which there is a linear relationship between input and output variables. These are used to predict continuous output variables, such as market trends, weather prediction, etc. Find the relationship between one dependent variable and a series of other changing variables. Some popular Regression algorithms are given below:

Simple Linear Regression Algorithm, Multivariate Regression Algorithm, Decision Tree Algorithm, Lasso Regression



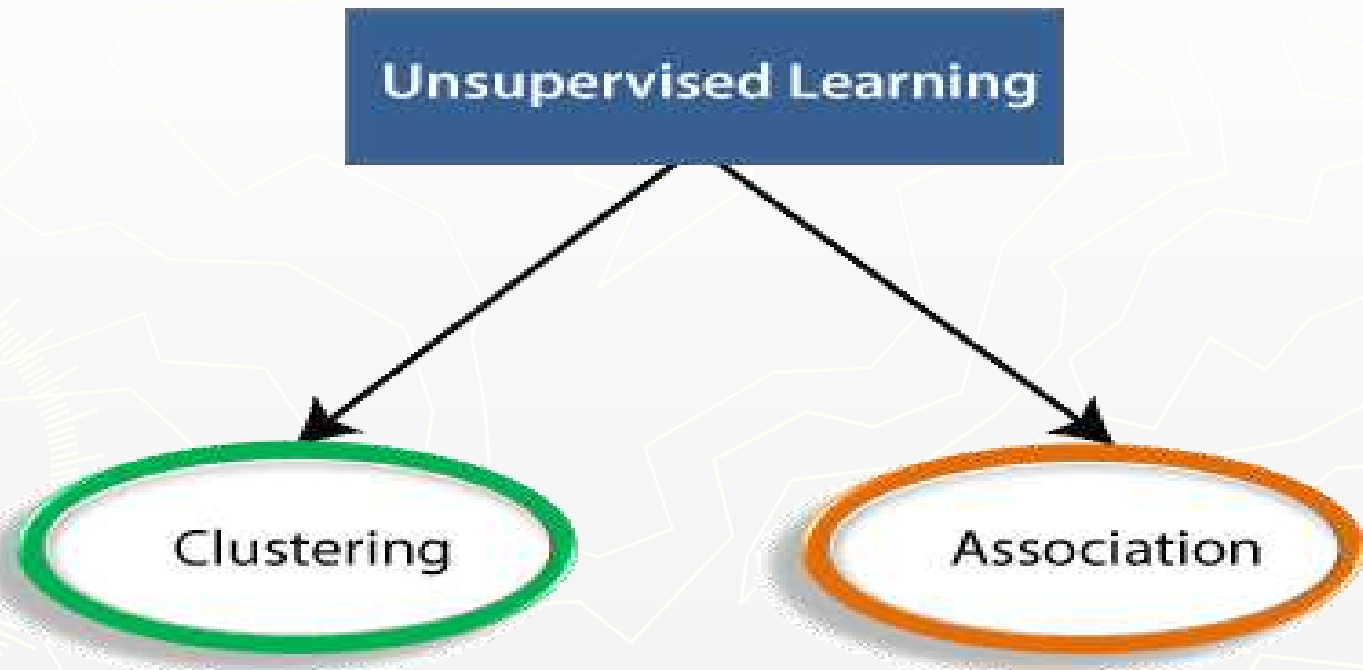
I DON'T TRUST LINEAR REGRESSIONS WHEN IT'S HARDER TO GUESS THE DIRECTION OF THE CORRELATION FROM THE SCATTER PLOT THAN TO FIND NEW CONSTELLATIONS ON IT.

Machine Learning Branches



Machine Learning Branches: Unsupervised Learning

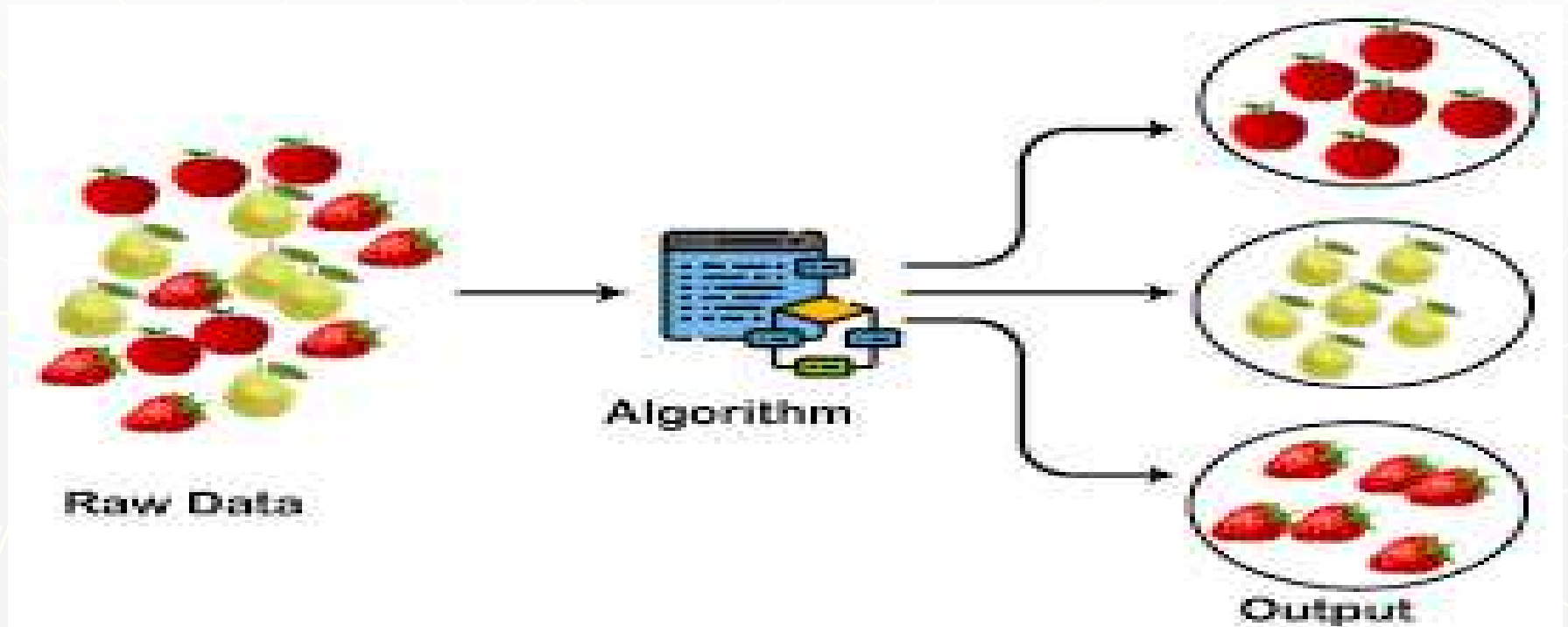
In **unsupervised learning** there are no labels available, insights are gained without* prior knowledge. The main aim of the unsupervised learning algorithm is to group or categories the unsorted dataset according to the similarities, patterns, and differences



* Usually some model parameters need to be set ahead of training.

Unsupervised Learning: Clustering

Clustering: The clustering technique is used when we want to find the inherent groups from the data. It is a way to group the objects into a cluster such that the objects with the most similarities remain in one group and have fewer or no similarities with the objects of other groups. An example of the clustering algorithm is grouping the customers by their purchasing behaviour. Some of the popular clustering algorithms are given below: K-Means Clustering algorithm, Mean-shift algorithm, DBSCAN Algorithm, Principal Component Analysis, Independent Component Analysis



Unsupervised Learning: Association

Association rule learning is an unsupervised learning technique, which finds interesting relations among variables within a large dataset. The main aim of this learning algorithm is to find the dependency of one data item on another data item and map those variables accordingly so that it can generate maximum profit. This algorithm is mainly applied in **Market Basket analysis**, **Web usage mining**, **continuous production**, etc.

Some popular algorithms of Association rule learning are **Apriori Algorithm**, **Eclat**, **FP-growth algorithm**.



Customer 1



Customer 2

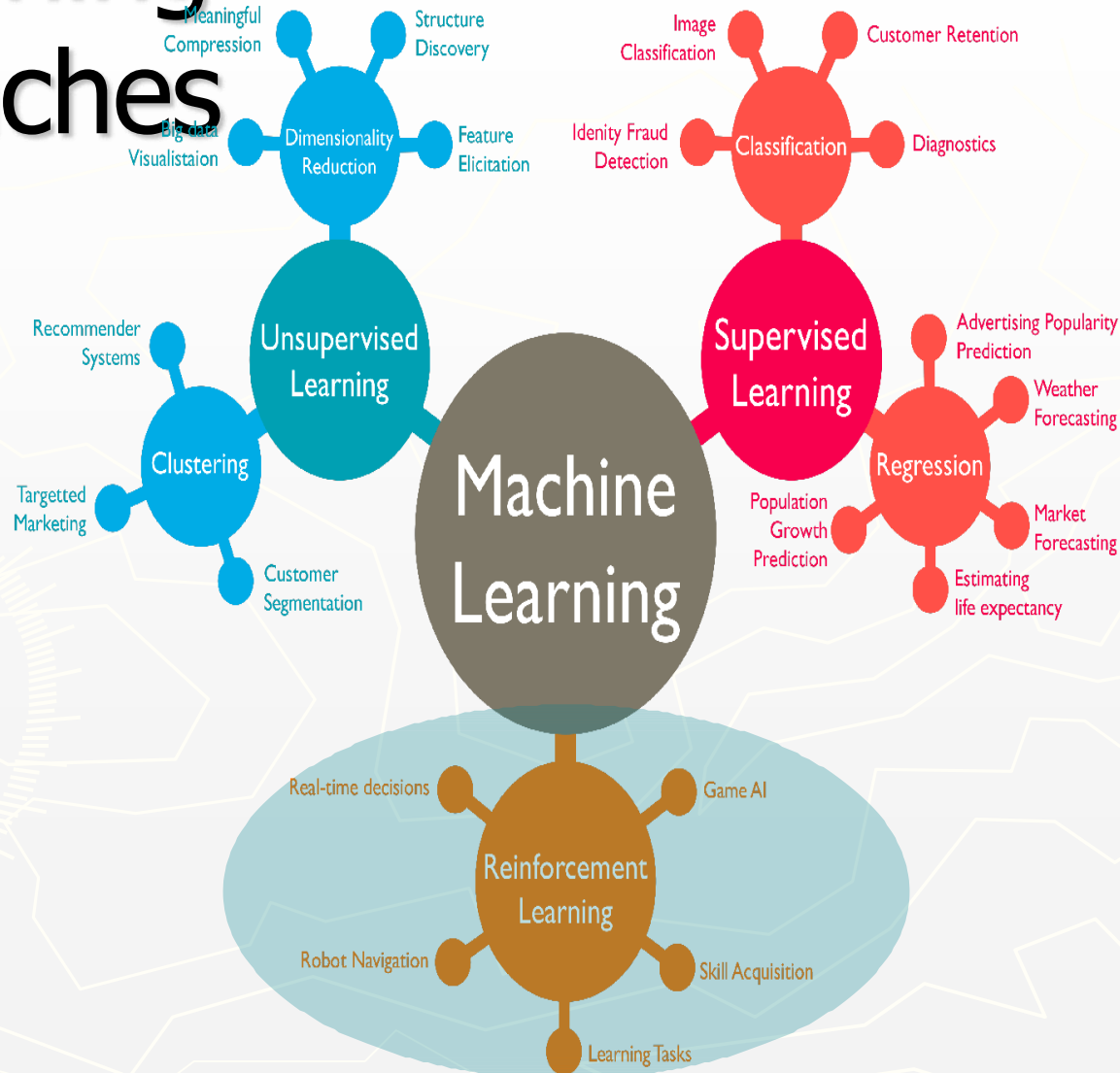


Customer 3



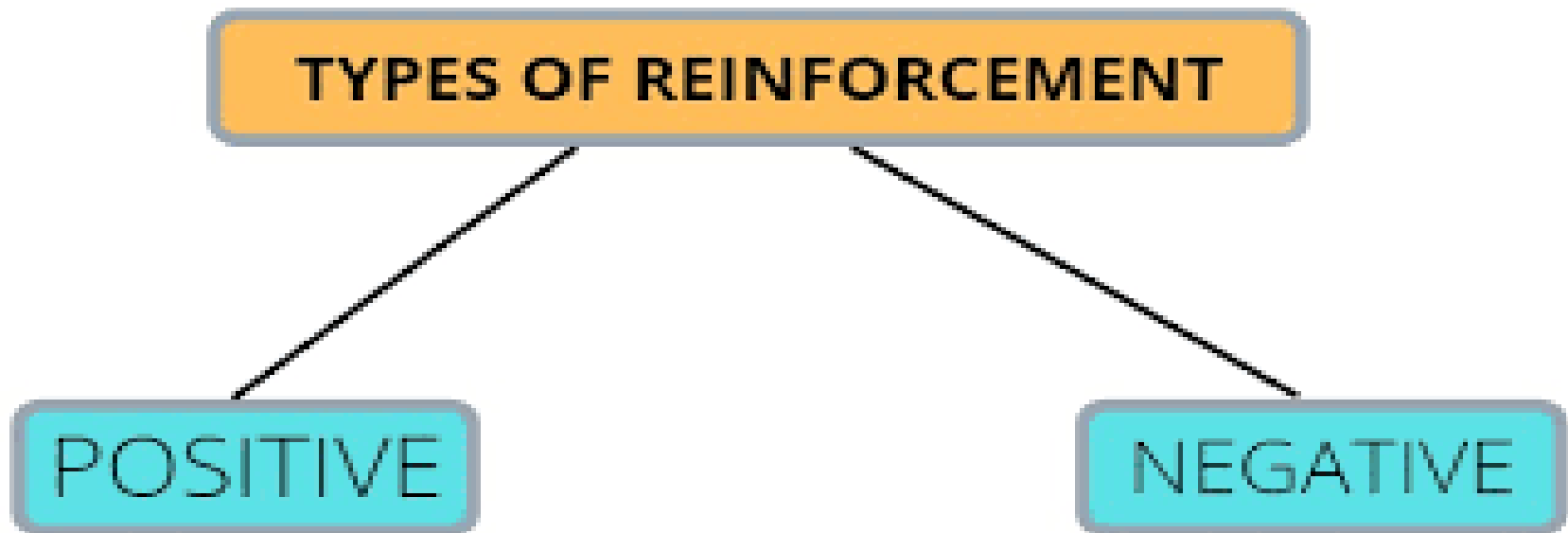
Customer n

Machine Learning Branches



Machine Learning Branches: Reinforcement Learning

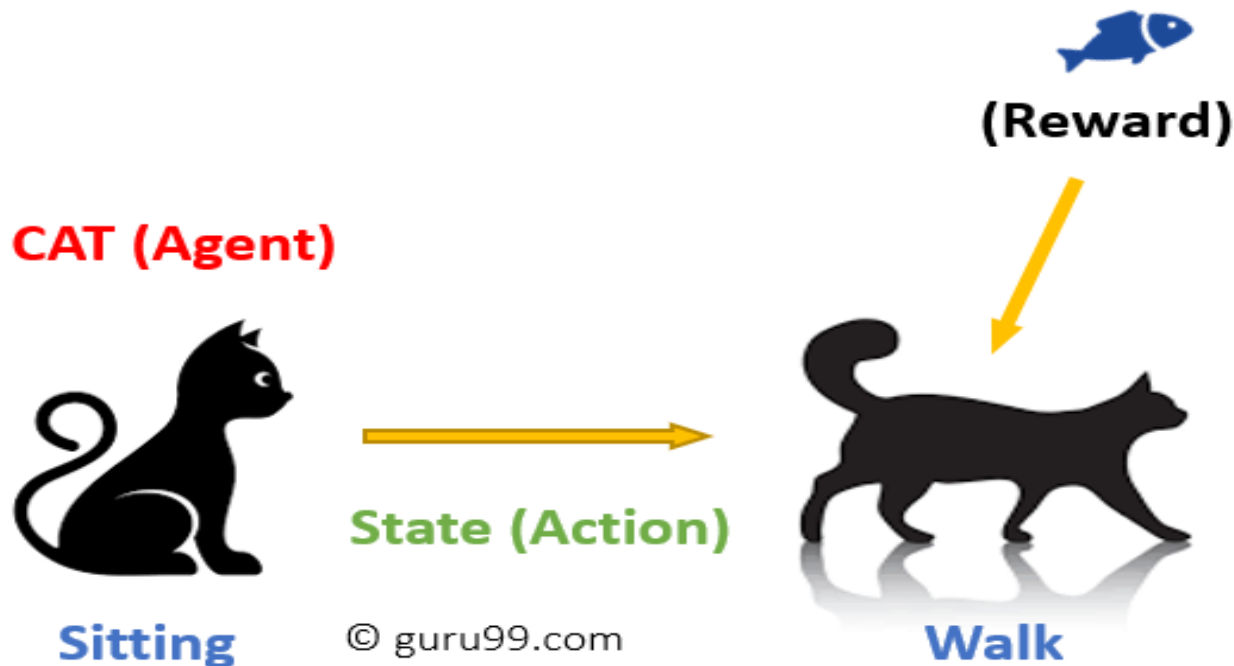
Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences, and improving its performance. Agent gets rewarded for each good action and get punished for each bad action; hence the goal of reinforcement learning agent is to maximize the rewards. Due to its way of working, reinforcement learning is employed in different fields such as **Game theory, Operation Research, Information theory, multi-agent systems.**



Reinforcement Learning: Positive

Positive Reinforcement Learning: Positive reinforcement learning specifies increasing the tendency that the required behaviour would occur again by adding something. It enhances the strength of the behaviour of the agent and positively impacts it.

House (environment)



Reinforcement Learning: Negative

Negative Reinforcement Learning: Negative reinforcement learning refers to the strategy of encouraging an agent to take actions that reduce or eliminate negative outcomes, shaping its behavior through the removal of undesired penalties.

Negative Reinforcement Learning



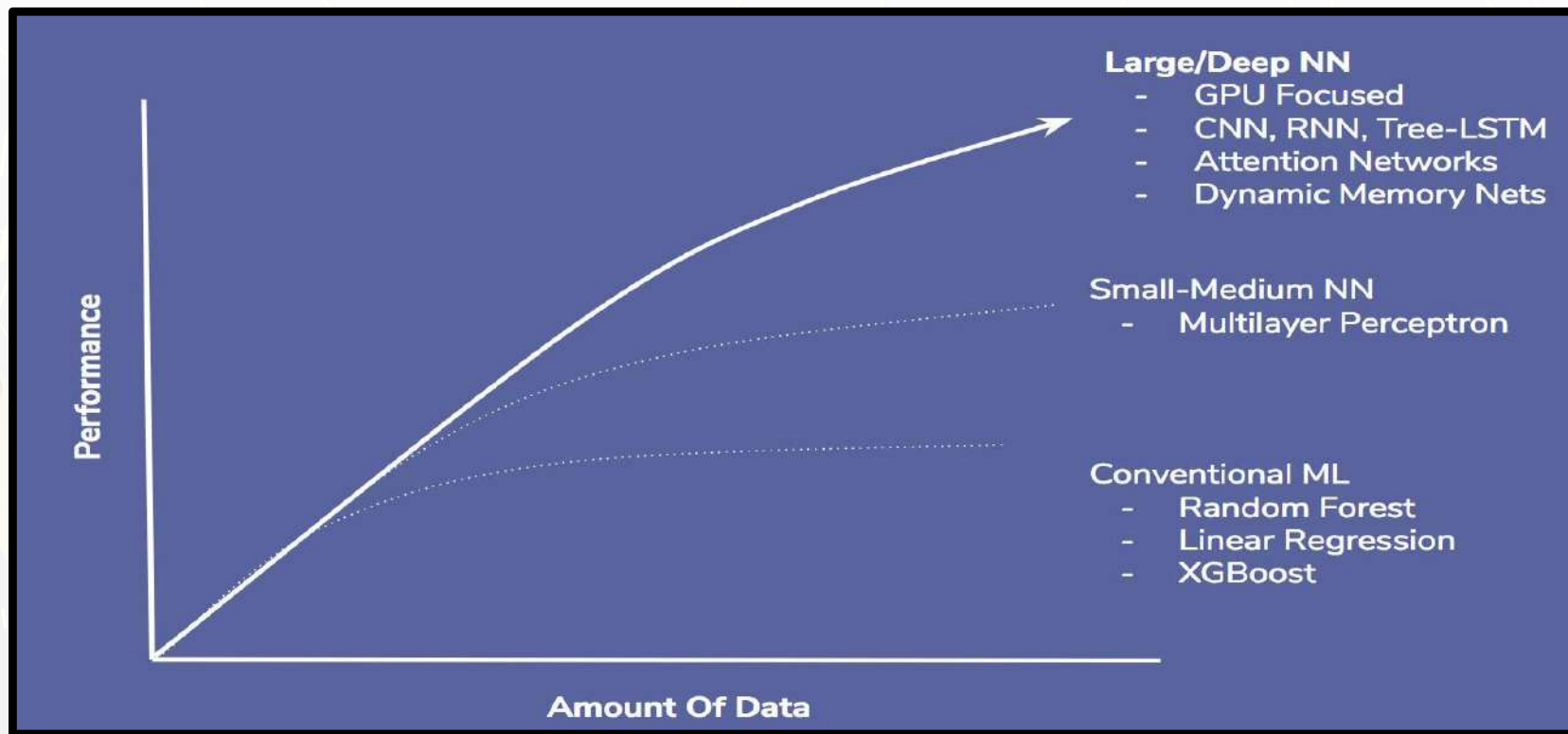
Behavior Strengthened by Removing an Unpleasant Stimulus

Part II: Deep Learning

Deep learning (DL) is the type of machine learning (ML) that resembles human brains, where it learns from data by using artificial neural networks. Just like human brains, these deep neural networks learn from real-life examples. In a few cases, it has surpassed human intelligence, just like Google's AlphaGo has defeated number one Go Player Ke Jie.

Why now? In recent years two things became available:

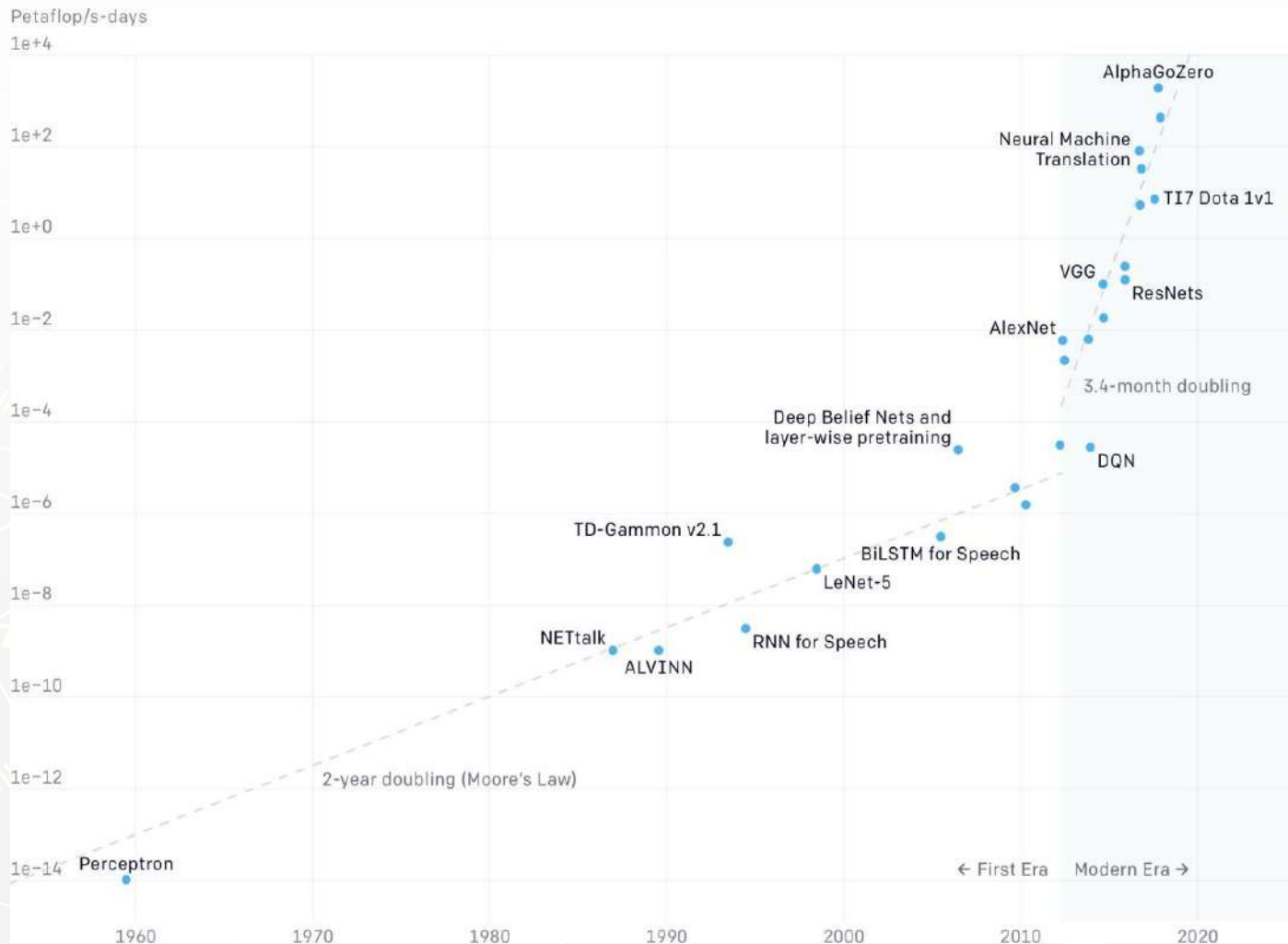
1. A lot of data



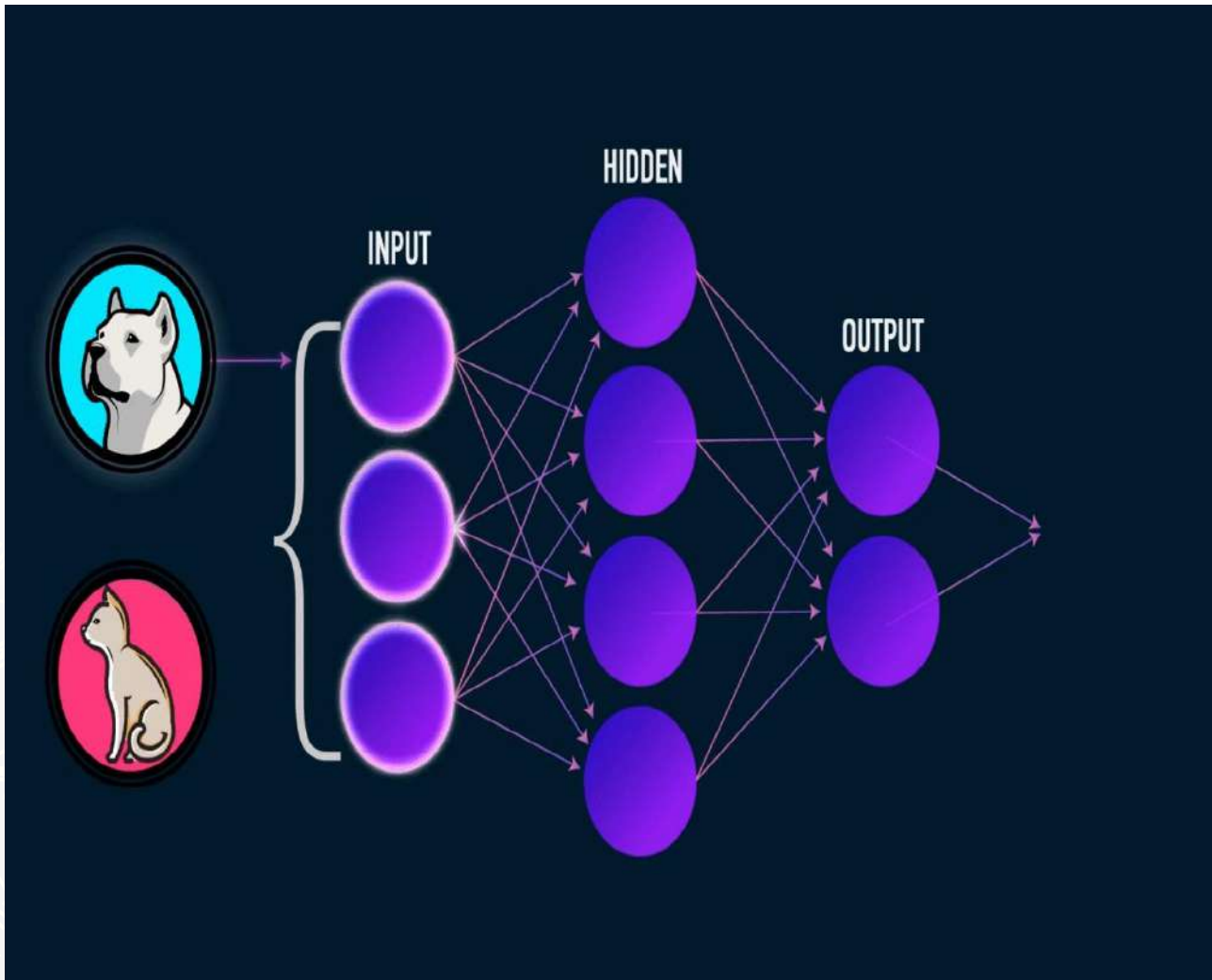
Part II: Deep Learning

Why now? In recent years two things became available:

2. *Necessary compute*



What is deep learning?

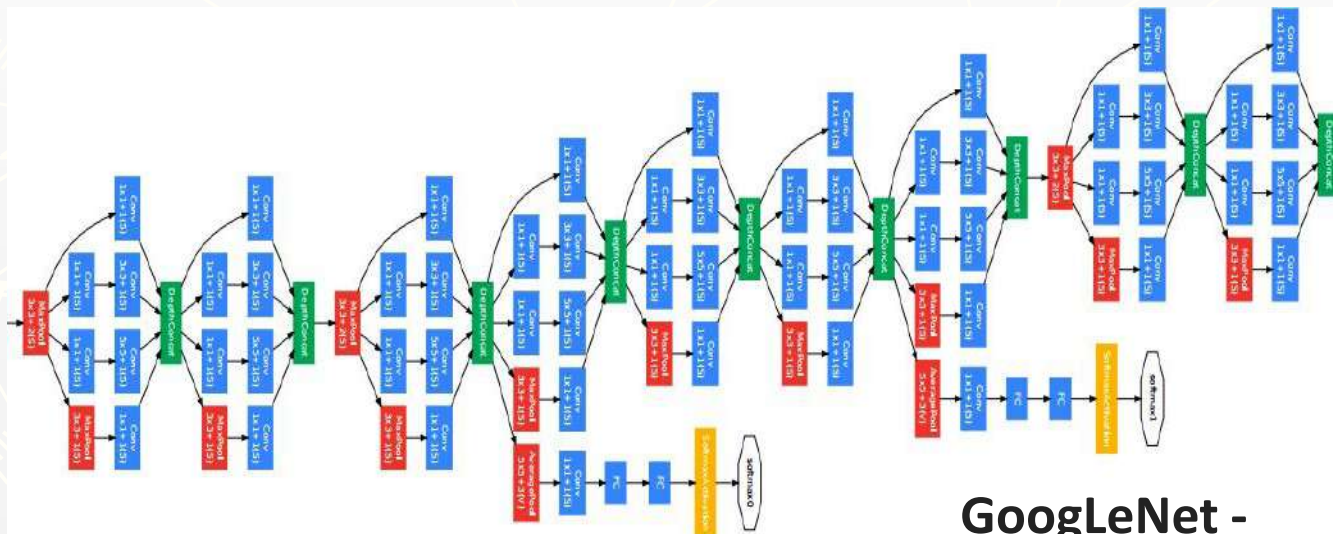


Rosenblatt -
1961

What is new in deep learning?

What is new (among other things) is a learning algorithm called **backpropagation** which allows to train **deep neural nets**.

State-of-the-art networks can have **over 200 layers!**



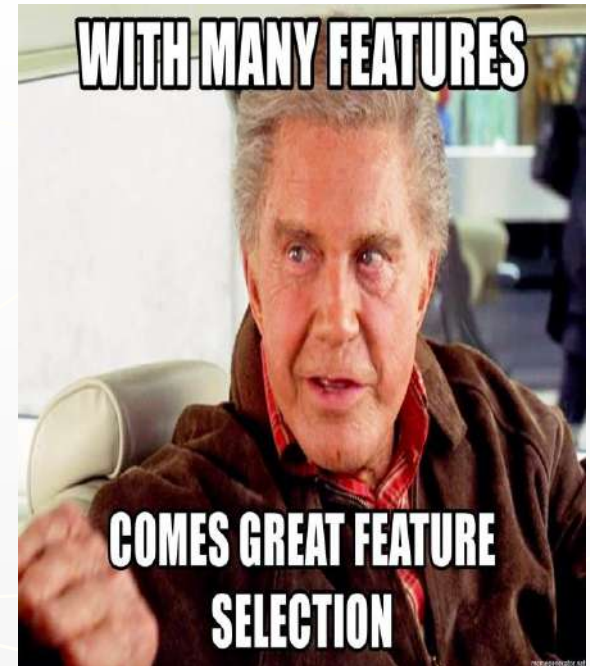
GoogLeNet -
2014

Difference classical ML vs. Deep Learning

Classical ML methods don't handle high dimensionality well.

⇒ **dimensionality reduction** & **feature selection**

Deep neural nets learn **compact representations** of data even in a high dimensionality/sparse setting - **no feature engineering** required!



Unsupervised Learning: Generative Adversarial Nets (GAN)



2014



2015



2016



2017



2018

NVIDIA

None of these images were taken in the real world!

Unsupervised Learning: Generative Adversarial Nets (GAN)



DeepMind - 2018

Which of these images was generated?

Unsupervised Learning: Language Generation

OpenAI, February 14 2019: Better Language Models and Their Implications

SYSTEM PROMPT
(HUMAN-WRITTEN)

In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

MODEL COMPLETION
(MACHINE-WRITTEN,
10 TRIES)

The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

Pérez and the others then ventured further into the valley. "By

Unsupervised Learning: Language Generation

OpenAI, February 14 2019: Better Language Models and Their Implications

www.talktotransformer.com



So why not use Deep Learning for everything?



- ▶ There are reasons why we don't only use DL:
- Necessary data no available
- Computational power not available
- Harder to interpret results
- Deep networks can be fooled:

